The Meteoritical Society

Meteoritics & Planetary Science 51, Nr 10, 1762–1794 (2016)
doi: 10.1111/maps.12710

Impacts into quartz sand: Crater formation, shock metamorphism, and ejecta distribution in laboratory experiments and numerical models

Kai WÜNNEMANN[1,*], Meng-Hua ZHU[1,2], and Dieter STÖFFLER[1]

[1]Museum für Naturkunde, Leibniz Institute for Evolution and Biodiversity Science, Invalidenstraße 43, 10115 Berlin, Germany
[2]Space Science Institute, Macau University of Science and Technology, Avenue Wailong, Taipa, Macau
[*]Corresponding author. E-mail: kai.wuennemann@mfn-berlin.de

(Received 09 June 2016; revision accepted 16 June 2016)

Abstract–We investigated the ejection mechanics by a complementary approach of cratering experiments, including the microscopic analysis of material sampled from these experiments, and 2-D numerical modeling of vertical impacts. The study is based on cratering experiments in quartz sand targets performed at the NASA Ames Vertical Gun Range. In these experiments, the preimpact location in the target and the final position of ejecta was determined by using color-coded sand and a catcher system for the ejecta. The results were compared with numerical simulations of the cratering and ejection process to validate the iSALE shock physics code. In turn the models provide further details on the ejection velocities and angles. We quantify the general assumption that ejecta thickness decreases with distance according to a power-law and that the relative proportion of shocked material in the ejecta increase with distance. We distinguish three types of shock metamorphic particles (1) melt particles, (2) shock lithified aggregates, and (3) shock-comminuted grains. The agreement between experiment and model was excellent, which provides confidence that the models can predict ejection angles, velocities, and the degree of shock loading of material expelled from a crater accurately if impact parameters such as impact velocity, impactor size, and gravity are varied beyond the experimental limitations. This study is relevant for a quantitative assessment of impact gardening on planetary surfaces and the evolution of regolith layers on atmosphereless bodies.

INTRODUCTION

Impact craters form through the displacement and ejection of material. Most of the debris expelled from a crater is deposited as a more or less continuous ejecta blanket (up to ~2 crater radii R) whose characteristics depend on many parameters, including the distance from the point of impact; target composition and properties; or, to some extent, the presence or absence of an atmosphere. The ejection process follows principles of shock physics, mechanics, and ballistics and is generally well understood (see e.g., Melosh 1989). Qualitatively, the excavation zone (the region inside the transient crater wherefrom material is expelled from the crater) can be approximated by a moat around the impact site with a depth (the excavation depth) of about 1/10 of the transient crater diameter. Except for the immediate range next to the crater rim, the amount of ejecta deposited in the vicinity of the crater decreases as a function of distance according to a power-law (e.g., Carlson and Jones 1965; McGetchin et al. 1973; Stöffler et al. 1975; Housen et al. 1983). Early ejecta derive from stratigraphically upper units close to the point of impact and are the fastest traveling the farthest. With increasing launch distance from the point of impact, the ejection velocity decreases. However, the shock compression varies for material that is ejecta at the same distance and as a consequence, material that has experienced different degrees of shock metamorphism including melting is eventually deposited at the same distance. With further increase in radial range, material is disrupted into debris, where the fragment size approximately correlates with decreasing stresses and strain rates. Ejecta that leave the crater toward the end of the formation of the transient crater are the largest

© The Meteoritical Society, 2016.

fragments that have experienced the lowest stresses and accelerations.

Our current knowledge of the quantitative distribution of ejecta is based on laboratory impact experiments; numerical modeling; and observations at a few, well-preserved terrestrial craters (e.g., the Ries crater, Hörz et al. 1983; Artemieva et al. 2013; Stöffler et al. 2013; Sturm et al. 2015). Impact experiments have been carried out primarily with targets consisting of loose material such as quartz sand or glass beads (e.g., Gault and Heitowit 1963; Gault et al. 1968; Schmidt and Housen 1987). A great deal of work also exists on the dimensional analysis of the ejection of material based on impact and explosion experiments (e.g., Andrews 1977; Housen et al. 1983; Housen and Holsapple 2011). More recent experiments use particle image velocimetry (PIV) to record the trajectories of individual particles from the point in time when they leave the crater (launch time and place) to their terminal position (Cintala et al. 1999; Heineck et al. 2002; Hermalyn and Schultz 2011), but they neither resolve the source area of the ejecta nor the shock conditions the material was exposed to, as individual grains were not marked and recovered after the experiments. In a pioneering study, Stöffler et al. (1975) addressed these shortcomings by using ejecta catchers and layers of differently colored sand to reconstruct the preimpact target location of ejecta and to analyze their degree of shock loading. However, since the PIV method was not available at that time, the determination of the launch velocity and angle to calculate the parabolic trajectories was not possible. The method also does not take into account that ejecta move along the surface after landing entraining local material into the eject deposit.

Complementary numerical models can provide the missing information on the ejection mechanics and shock conditions for impact experiments performed in the 1970s and further allow systematic parametric studies on the effect of target properties such as porosity and strength on ejecta distribution (Collins and Wünnemann 2007). In some studies, the effect of an atmosphere and resulting deviations from ballistic trajectories and the interaction with the transient ejecta plume have been investigated (e.g., Artemieva and Morgan 2009; Artemieva et al. 2013); however, only very few modeling studies of the ballistic ejection in comparison to experiments (Wada et al. 2006; Collins and Wünnemann 2007) and observations at natural craters (Collins et al. 2008; Artemieva et al. 2013) have been carried out so far and a systematic validation study is lacking.

Here, we provide a new attempt to compare numerical models of the excavation and ejection of material during an impact event with data obtained from a suite of laboratory impact experiments. The latter were carried out in 1972, but have been analyzed and published only in parts (Stöffler et al. 1975). The goal is to determine ejection velocity and angle and the amount of ejecta as a function of distance from the preimpact location. Also, the optical and SEM analysis of the shock history experienced by the experimental ejecta from different-colored source regions were conducted, thus providing new and unique constraints on modeling the shock-decay within the sand targets. The overall objectives were to compare experimental and numerical results, hopefully validating the latter. Thoroughly validated models will allow for investigating —with increased confidence—the effects of a wide range of parameters such as porosity, strength, and lithology on ejecta distribution and will extend beyond laboratory experiments the range of scenarios that can be modeled, including large-scale impacts. Detailed quantitative understanding of the relationship between the preimpact location of matter in the target and the ejection distance and shock wave exposure as a function of crater size and target properties is essential to understand the lateral and vertical redistribution of planetary surface materials and their shock histories.

First, we describe the impact experiments and the numerical methods we used to simulate crater formation and to quantify the ejection process. Next, we present the macroscopic results from modeling and experiments in terms of crater size, ejecta distribution and their place of origin, and the attenuation of shock pressure in the target. Subsequently, we describe the observed stages of shock metamorphism in the ejecta samples and provide data on the distribution of variably shocked quartz sand and its grain size as a function of distance. Finally, we summarize the comparison between experiments and modeling and draw conclusions with regard to the reconstruction of the impact ejecta record on planetary surfaces.

EXPERIMENTAL TECHNIQUES

The experimental data are based on hypervelocity impact cratering experiments performed at the Vertical Gun Range of the NASA Ames Research Center, Moffett Field, California (http://www.nasa.gov/centers/ames/thermophysics-facilities/ranges or http://thermophysics.arc.nasa.gov). Projectiles consisted of plastic (LEXAN) cylinders (density: 1.2 g cm^{-3}) and aluminum spheres (density: 2.8 g cm^{-3}) with a mass of 0.30 g and 0.38 g, respectively. They were fired vertically into targets of quartz sand with velocities ranging from 5.86 km s^{-1} to 6.90 km s^{-1}. A total of 15 experiments were carried out. Results of only six experiments with

Fig. 1. Impact chamber (diameter: 2.2 m) of the NASA Ames Vertical Gun Range: a) Quartz sand target for experiment No. 387 placed in the central target bucket (diameter: 60 cm), which is surrounded by horizontal ejecta catchers made of honeycombed cardboard ("Hexcell") and bordered by vertical catchers (only the lower four catchers shown); the latter are positioned at a lateral distance of 110 cm and are subdivided into 5 shells covering angles of 0–21.8°, 21.8–32.9°, 32.9–41.2°, 41.2–46.9°, and 46.9–51° relative to the ground plane; b) crater of experiment No. 418 with catchers partly removed and vertical catchers with angles relative to the ground plane of 0–10°, 10–20°, 20–30°, 30–40°, and 40–50°; aluminum foil at the wall was used for recording the upper limit of ejection angles of high-speed ejecta; c) crater of experiment No. 387 surrounded by Hexcell catchers; d) same crater after removal of catchers displaying ejected sand on the ground as well as the circular and radial markers on the ground for sampling specific annuli in different sectors of 45°; e) crater of experiment No. 418 showing the sampling method for the ejecta blanket inside the bucket (several annuli of 2–3 cm diameter sampled in a 45° sector exposing ground zero marked with a thin layer of black lacquer; scale bar = 10 cm); part of the Hexcell at lower left; f) same crater with partly removed Hexcell catcher (lower left) leaving the ground markers for sampling visible; width of outer annular markers is 15 cm. Note that the catcher arrangement shown in (a) was used for Series I experiment No. 387 and for all nine experiments of Series II, whereas the arrangement shown in (b) was applied for Series I experiments 417, 418, and 421 (see Tables 1a–c, 3a, and 3b).

LEXAN projectiles (Series I) were published after the project campaign was completed (Stöffler et al. 1975). The remaining experiments with aluminum projectiles (Series II) that were so far only made available in conference abstracts (Stöffler et al. 1980a, 1980b; Jammes et al. 1983), were reinvestigated here. Both old and new observations were used as constraints for the numerical models.

Experimental Set-Up

The targets were embedded into a 60 cm wide and 20.5 cm deep cylindrical metal bucket placed in the center of the 2.2 m wide impact chamber (Fig. 1) that was evacuated to an ambient pressure of 67–400 Pa (0.5–3 mm Hg). Ejecta from the experimentally produced craters with diameters between 29.5 cm and 33.6 cm were collected at various radial distances r from the point of impact ranging from r = 18 and 16 cm to r = 110 cm and 105 cm (~1.1 to ~7 and ~6 crater radii) for experiment Series I and Series II, respectively. In addition, ejecta were sampled in vertical catchers mounted to the chamber wall at r = 110 and 105 cm, respectively. The sampling within the target bucket was made possible by spraying the target surface with a thin layer of black lacquer before the projectile was fired (see also Fig. 4) in order to mark the preimpact target surface and avoid mixing of ejecta with the target surface. The sampling outside the target bucket was accomplished by placing catchers made of "Hexcell" (trade name), a honey-combed cardboard, on the reference ground plane and inside aluminum frames at the chamber wall (vertical catchers) as shown in Fig. 1. The catchers were used to avoid lateral displacement of particles upon landing. The ground plane below the catchers was marked with circular and radial lines indicating the annuli (5–15 cm in width) and radial 90° sectors (A, B, C, D) from which the samples were collected after removal of the Hexcell cardboard (Fig. 2). Inside the target bucket ejecta were also collected from 2–3 cm wide annuli, which were cut into the ejecta blanket by a special device after the experiment (Fig. 1).

Projectiles and Impact Velocities

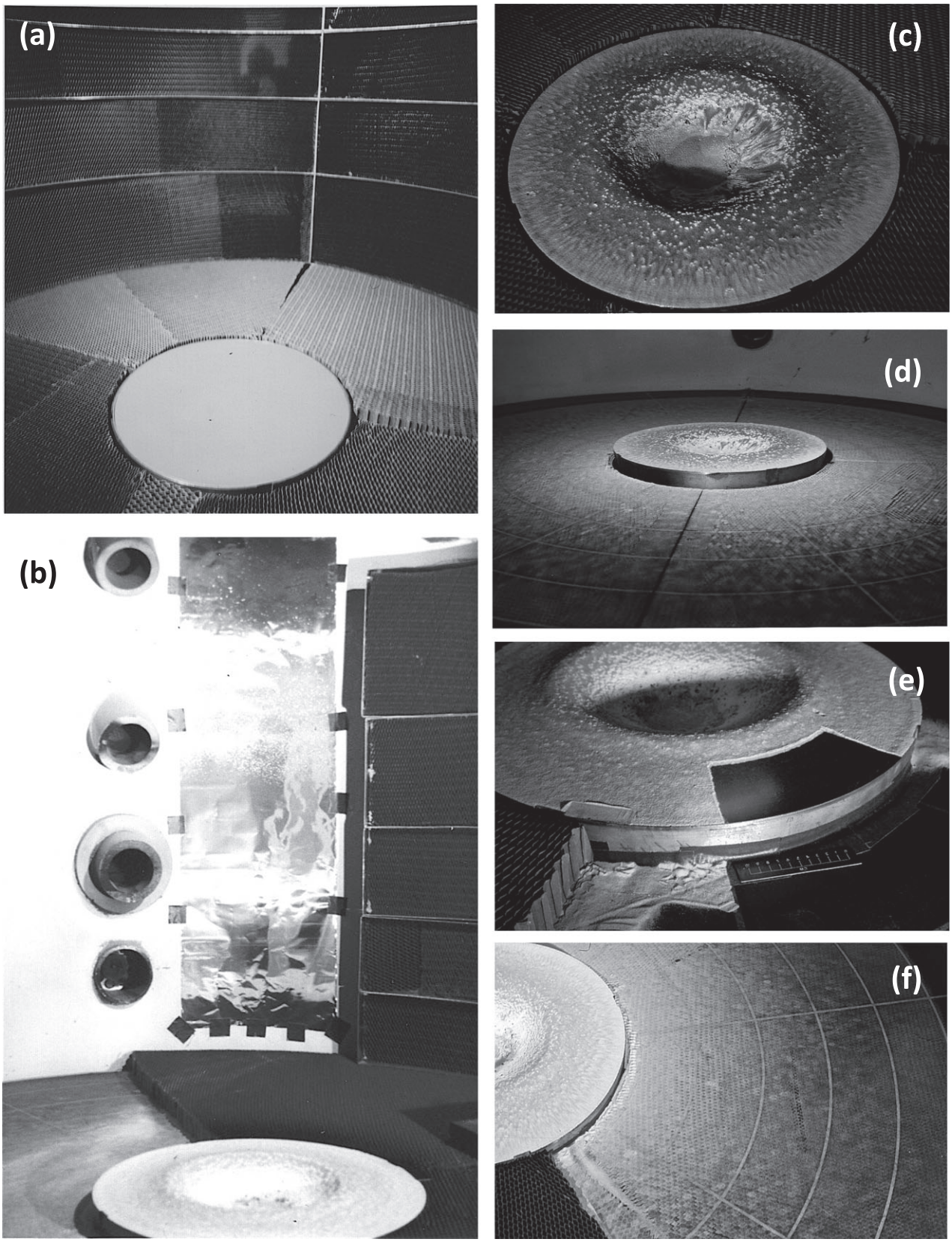
For the Series I experiments, cylindrical projectiles made of thermoplastic polycarbonate (trade name LEXAN) measuring 0.79 cm in diameter and 0.51 cm in length (density = 1.2 g cm^{-3}) were used. Their mass varied from 0.268 g to 0.311 g (Table 1a and Stöffler et al. 1975). Impact velocities varied from 5.87 km s^{-1} to 6.90 km s^{-1}. For the Series II experiments, aluminum spheres with a diameter of 0.636 cm and a mass of 0.38 g (density = 2.8 g cm^{-3}) were accelerated to 5.86 km s^{-1} to 6.46 km s^{-1} (Table 1b). The spheres were made from Duralumin, an Al-alloy with 94.9 wt% Al, 4.20 wt% Cu, 0.73 wt% Mn, 0.61 wt% Mg, 0.67 wt% Fe, and 0.42 wt% Si (see also Hamann et al. 2016).

Target Configuration

For all experiments of Series I and II (see Tables 1a–c), commercial quartz sand with a grain size of 0.063–0.25 mm was used which contained >99% quartz. The exact grain size distribution by mass is given in Table 1c. For color coding, the sand was dyed with special liquids of different color, dried at moderate temperature, and subsequently sieved through a 350 sieve with a mesh size of 0.5 mm. Due to the irregular shape of the sand grains, they were not entirely covered with dye. The bulk density of the sand of different color was determined for the Series I experiments, and ranged from 1380 kg m^{-3} to 1550 kg m^{-3} compared to 1580 kg m^{-3} for plain sand. The coloring of sand resulted in a grain size distribution of the sand that is slightly different from the plain sand as documented in Stöffler et al. (1975) and Table 1c.

For Series I experiments, four types of target setups were used (Figs. 3a and 4; Tables 2a and 2b):

Type 1: plain sand


(a)
(b)
(c)
(d)
(e)
(f)

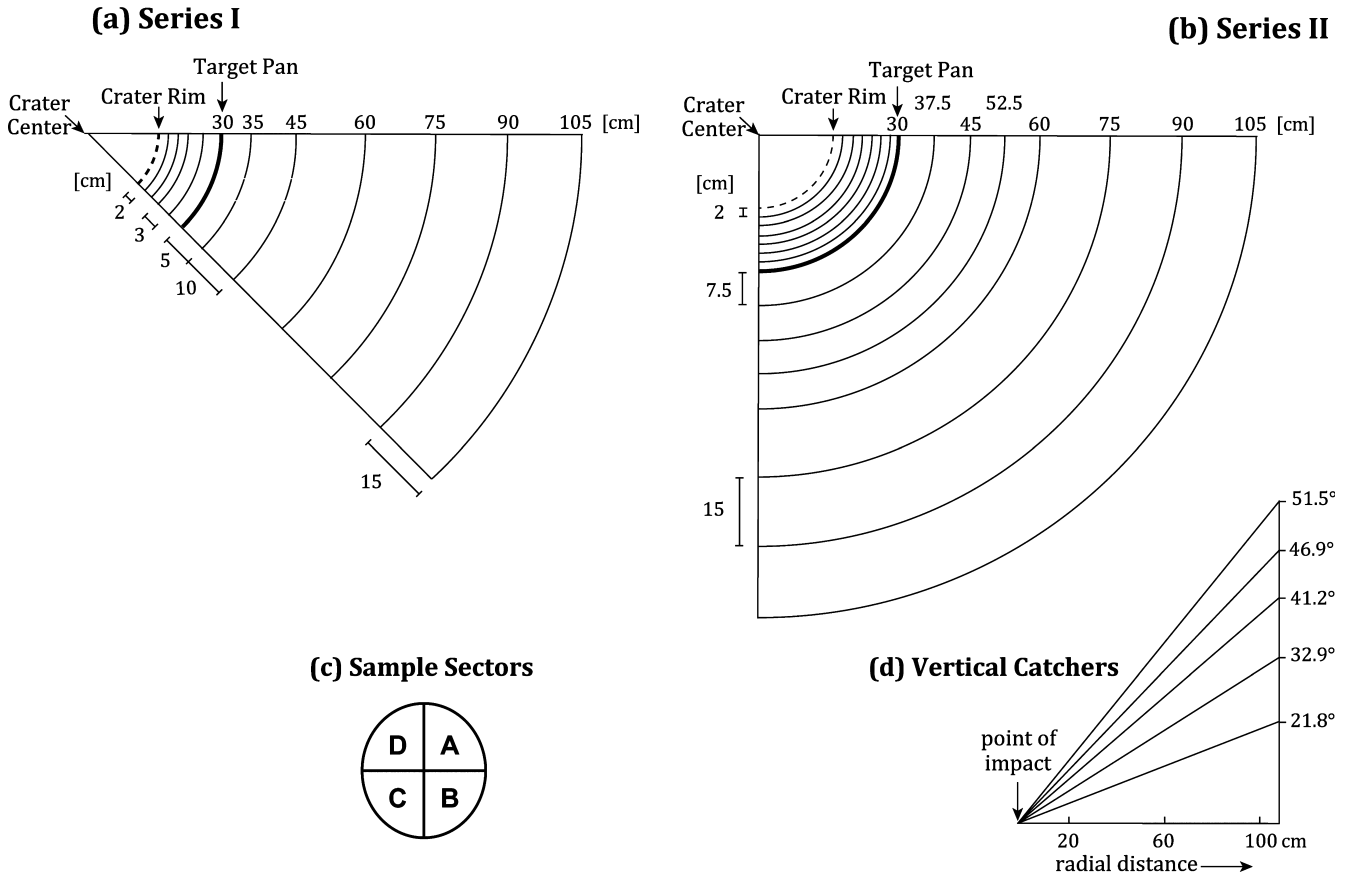


Fig. 2. Sampling plans for cratering experiments of Series I (a) and Series II (b). Location of sectors A, B, C, D, two of which were sampled in Series II, is indicated in (c); Location of vertical catchers is shown in (d). For further details, see text and Tables 1a–c and 3a–c.

Type 2: sequence of five, 9 mm thick layers of differently colored sand on top of plain sand (Fig. 3a, top-left)

Type 3: one 11 mm thick top layer of colored sand on top of a sequence of four, 9 mm thick layers of differently colored sand on top of plain sand (Fig. 3a, top-right)

Type 4: one 7 mm thick layer consisting of four consecutive annuli of differently colored sand on top of plain sand (Fig. 3a, bottom)

In one experiment of type 2 (No. 420) all sand layers were mixed with 2 wt% of epoxy powder allowing solidification by baking for subsequent cross cutting of the entire crater (Fig. 4). The "nested" crater morphology in experiment No. 420 is most likely due to the change in rheological properties of the colored sand relative to the plain sand and the epoxy powder and did not occur in most other experiments. All craters of Series II are rather cone-shaped (see the Crater Morphometry section).

For Series II, also four different types of targets were used (Fig. 3b; Table 2b):

Type 5: one 9 mm thick layer consisting of six consecutive rings of differently colored sand on top of plain sand (Fig. 3b, Nos. 529 and 549)

Type 6: same as type 5 but colored layer at 9 mm depth with plain sand on top and below (Fig. 3b, Nos. 530 and 548)

Type 7: same as type 6 but colored layer at 18 mm depth (Fig. 3b, Nos. 531 and 550)

Type 8: same as type 6 but colored layer with only four consecutive rings at 27 mm depth (Fig. 3b, Nos. 532, 533, and 534).

The structures and color coding of all 8 target types of both Series I and II, are summarized in Tables 2a and 2b and Fig. 3.

Quantitative Techniques for the Analyses of Craters and Their Ejecta

The first step of the postimpact analyses was the photographic documentation of the crater and the ejecta blanket after removal of the Hexcell cardboard catchers on the ground plane (Figs. 1 and 2). In addition, the

Table 1a. Experimental conditions and crater dimensions for experiments with LEXAN projectiles with a density of 1.2 g cm^{-3} (Series I) impacted into targets of quartz sand (grain size: 0.063 to 0.250 mm) consisting of plain sand or of 7, 9, or 11 mm thick layers of colored sand (Table 2a); projectiles are cylinders of polycarbonate (LEXAN) with a diameter of 0.79 cm and a length of 0.51 cm and an average mass of 0.300 g; crater dimensions were determined from crater profiles (see Fig. 5); D_a = apparent crater diameter at reference surface, D_r = crater diameter at rim, d = crater depth measured from reference surface, h = rim height measured from reference surface, Volume = crater volume, Mass = crater mass (the crater mass corresponds to the mass that was originally located inside the crater); see also table 1 in Stöffler et al. (1975).

Experiment No.[a]	Impact velocity (km s^{-1})	Projectile mass (g)	Kinetic energy (10^3 J)	D_r (cm)	D_a (cm)	d (cm)	h (cm)	Volume (cm^3)	Mass (g)
387	6.85	0.268	6.29	30.5	–	6.0	0.8	–	–
390	–	0.304	–	–	–	–	–	1446	2183
392	–	0.311	–	–	–	–	–	1461	2206
416	5.87	0.300	5.17	29.5	–	6.55	–	1039	1568
417	6.77	0.306	7.01	31.7	–	7.45	1.0	1236	1866
418	6.43	0.304	6.28	32.8	27.15	5.8	0.8	1467	2082
420	6.76	0.306	6.99	31.7	24.75	6.45	0.9	1383	1963
421	6.90	0.302	7.18	31.5	–	5.9	1.0	1167	1762
Averages	6.60	0.300	6.49	31.28	–	6.36	0.9		

[a]Experiments 390–417 are with targets of plain sand and experiments 387 and 418–420 with targets with five layers of differently colored sand; experiment 421 has a target with a top layer consisting of four torus rings of differently colored sand (for details see Table 2a).

Table 1b. Experimental conditions and crater dimensions for experiments with aluminum projectiles with a density of 2.8 g cm^{-3} (Series II) impacted into targets of quartz sand (grain size: 0.063–0.250 mm); targets contain one 9 mm thick layer with rings of differently colored sand at various depths (see Table 2b); projectiles are aluminum spheres with a diameter of 0.636 cm and a mass of 0.38 g; crater dimensions were determined from crater profiles (see Fig. 5); D_a = apparent crater diameter at reference surface, D_r = crater diameter at rim, d = crater depth measured from reference surface, h = rim height measured from reference surface; Volume = crater volume, Mass = crater mass (the crater mass corresponds to the mass that was originally located inside the crater).

Experiment No.	Impact velocity (km s^{-1})	Kinetic energy (10^3 J)	D_a (cm)	D_r (cm)	d (cm)	h (cm)	Volume (cm^3)	Mass (g)
529	6.44	7.8	26.85	32.8	5.8	0.9	1427	2140.5
530	6.46	7.85	26.3	33.1	6.0	0.9	1264	1896
531	6.17	7.16	27.4	32.8	6.4	0.8	1397	2095.5
532	6.24	7.32	27.05	32.6	5.3	0.9	1416	2124
549	5.86	6.46	26.85	31.95	6.1	0.7	1343	2014.5
548	5.99	6.75	26.4	31.85	6.4	0.7	1168	1752
550	5.97	6.7	28.0	32.4	6.3	0.7	1398	2097
534	6.32	7.51	28.3	33.65	7.4	0.8	1396	2094
533	6.11	7.02	28.35	32.6	6.5	0.7	1583	2374.5
Averages	6.17	7.17	27.28	32.64	6.24	0.79		

shape of the craters was determined with a mechanical profilometer device as shown in Fig. 6a. Several parameters were derived from the crater profiles such as apparent crater diameter D_a, crater rim diameter D_r, crater depth d, and height of crater rim h, volume and mass of the crater (Tables 1a–c). The crater mass M_c corresponds to the mass that was originally located inside the crater. The ejecta mass M_e is the total mass of ejected material and corresponds approximately to the volume of the excavation zone times the density of the target. The displaced mass is then given by $M_d = M_c - M_e$. The measured crater dimensions do not exactly correspond to those of the transient crater as modest gravity collapse occurred by slumping along over-steepened crater walls. From a series of unpublished recent impact cratering experiments in quartz sand by Wünnemann and Zhu using quarterspace targets and high-speed cameras that documented the evolving crater, we determined a ratio between final and transient crater diameter of $D_a/D_t = 1.08$, where D_t is the transient crater diameter measured at preimpact surface level. Ormö et al. (2015) found a compatible ratio of 1.1

Table 1c. Grain size distribution of plain and colored quartz sand used in the cratering experiments of Series I and II (Tables 1a and 1b); note the slight shift to larger grain sizes induced by the coloring process, which caused some grain sticking.

Grain size range (mm)	Type of target sand (wt%) Plain	Red	Yellow
<0.063	5.0	0	0
0.063–0.125	36.9	27.0	30.0
0.125–0.250	57.3	69.0	67.5
0.250–0.500	0.8	4.0	2.5

in similar cratering experiments in sand. Note, for natural simple craters on earth, Melosh (1989; page 129, Eq. 8.2.1) proposes a ratio of $D_a/D_t = 1.19$ for craters that were filled to half of the transient crater depth with slump material. The more pronounced collapse in natural events in comparison to the laboratory cratering experiments, where only little material slumps back into the crater is a result of the relative importance of gravity at different dimensional scales.

As described above, the ejecta were quantitatively sampled on the reference plane and in five levels of vertical catchers (Figs. 1 and 2, Tables 3a–c). For Series I experiments, samples were taken from concentric annuli of 2 and 3 cm radial width on the target bucket (Annuli 2–6 in Tables 3a–c) and from five annuli of 5, 10, and 15 cm width beyond the target bucket (Annuli 7–12), and from five vertical catchers up to an angle of 51.5° from the ground reference plane. For Series II experiments, samples were taken from six annuli of 2 cm width on the target bucket (Annuli 1–6), from four annuli of 7.5 cm width and four annuli of 15 cm width on the ground plane (Annuli 7–12), and from five vertical catchers up to an angle of 51.5°. The annuli were divided into 90° sectors (A, B, C, D) from which the samples were taken separately. The exact sampling plans are shown in Fig. 2 and documented in Tables 3a–c. Samples of Series I were taken from two 45° sectors from 11 consecutive annuli on the ground plane and five levels of vertical catchers (total of 32 samples; except experiment No. 387 for which 128 samples were taken). For the experiments of Series II two opposite or adjacent sectors (A, B, C, D) were completely sampled: 14 consecutive annuli on the ground plane and five levels of the vertical catchers (Fig. 2, Tables 3a–c). Due to some deviations of the impact points from the geometric center of the target bucket and the concentric rings of colored sand (see below), the outermost annuli on the target bucket was incomplete in most cases and, for this reason, was not sampled in Series II experiments. Depending on the size of the craters, also the innermost annuli could not always be sampled (Tables 3a–c).

Samples were also taken from inside the craters for both Series I and II experiments. An uppermost layer of the crater floors and walls was taken off a spoon to recover the fraction of the relatively large melt and shock lithified particles as quantitatively as possible using a 0.3 mm or 0.5 mm sieve for Series I and Series II experiments, respectively. From the <0.3/<0.5 mm fractions, the grain size distribution was determined for the allochthonous material inside the crater from which the mass fraction of shock-comminuted quartz (<0.063 mm or 0.125 mm for three experiments of Series II) was obtained. Because of the intrinsic uncertainty in distinguishing between allochthonous and "autochthonous" crater material, the data have relatively large errors.

For all samples defined above the following parameters were determined (1) total mass, (2) color distribution by counting 1000 grains under the stereomicroscope to determine their source area, (3) grain size distribution, (4) type and mass of shocked particles (melt particles and shock lithified particles (see below), and (5) their colors to determine their source area. In addition, the microscopic texture and the component phases of shocked particles were analyzed using the polarizing microscope and the scanning electron microscope.

For the Series I experiments 387 and 418 (Fig. 3a), the radial mass distribution of material from colored layers of variable preimpact depth was determined. For experiment 421 (Fig. 3a), the radial mass distribution of material from the 7 mm top layer was analyzed as function of its preimpact radial distance from the point of impact. Results of these measurements have been presented in detail previously (Stöffler et al. 1975; figs. 4, 5, 9, 10, and 18–20). In the case of Series II experiments, the color coding of the ejecta enables the reconstruction of depth and radial range where a specific sand grain was originally located. Combining different experiments with the colored source annuli located at different depths (see Fig. 3b; e.g., subseries 529, 530, 531, 532) of otherwise identical experiments allows for the 3-D reconstruction of the cratering flow field and the decaying shock wave. This ideal holds only partially, as there were slight variations in impact velocity and, as a consequence, variations in crater diameter. Additionally, the mass distribution of the ejecta was affected by some deviation of the point of impact from the geometric center of the target bucket and thus the center of the annuli of colored sand. This somewhat wanting aiming accuracy arises from the inaccuracies in the alignment of the gun axis and the target bucket and/or from the uncontrollable effects of sabot separation causing the projectile to veer off its

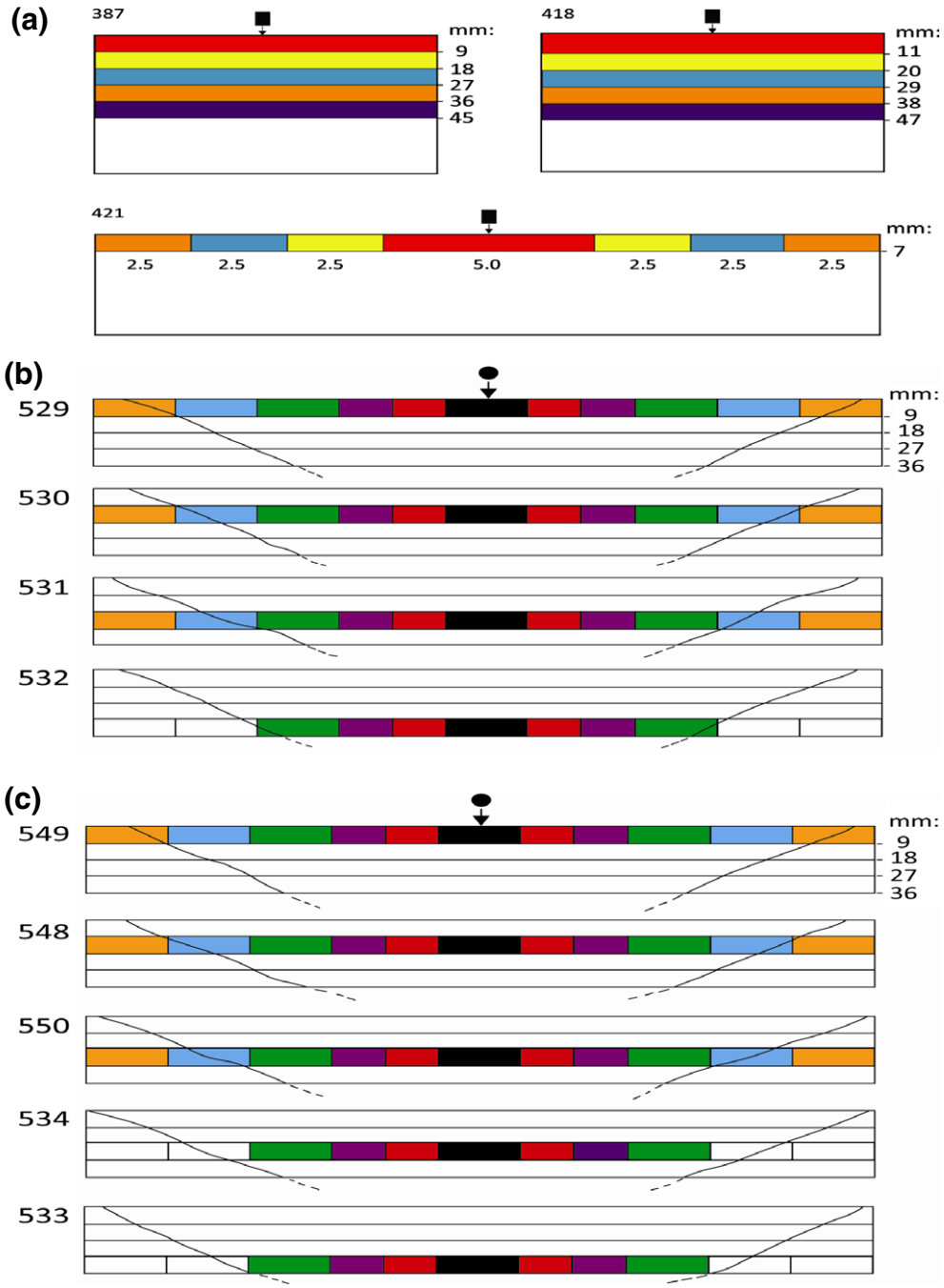


Fig. 3. Target setup: a) Color-coded target types used for Series I experiments 387, 418, and 421; top: vertical sequence of differently colored sand; bottom: top layer with 5 cm and 2.5 cm wide rings of differently colored sand (see also text and Table 2a); note that additional experiments were made in plain sand; arrow indicates the point of impact; black rectangle represents the projectile. b) Target types used for Series II experiments with 2 cm and 3 cm wide rings of differently colored sand at various depths for experiments Nos. 529–532 and (c) for experiments Nos. 533–534 and 548–550. Black sphere represents aluminum projectile; arrow indicates the point of impact; see also Table 2b; the inclined black lines indicate the crater profiles determined after impact.

nominal path. Numerous shots at the Ames Vertical Gun Range show that in vertical shots, the impact point of a projectile with a diameter of 0.63 cm (1/4 inch) hit the target aim point within a circle of 1.8 cm which we consider as the maximum error to hit the geometric center of the target. At experiment No. 387, for which all eight 45° sectors were sampled, we observed a standard deviation of the ejecta mass

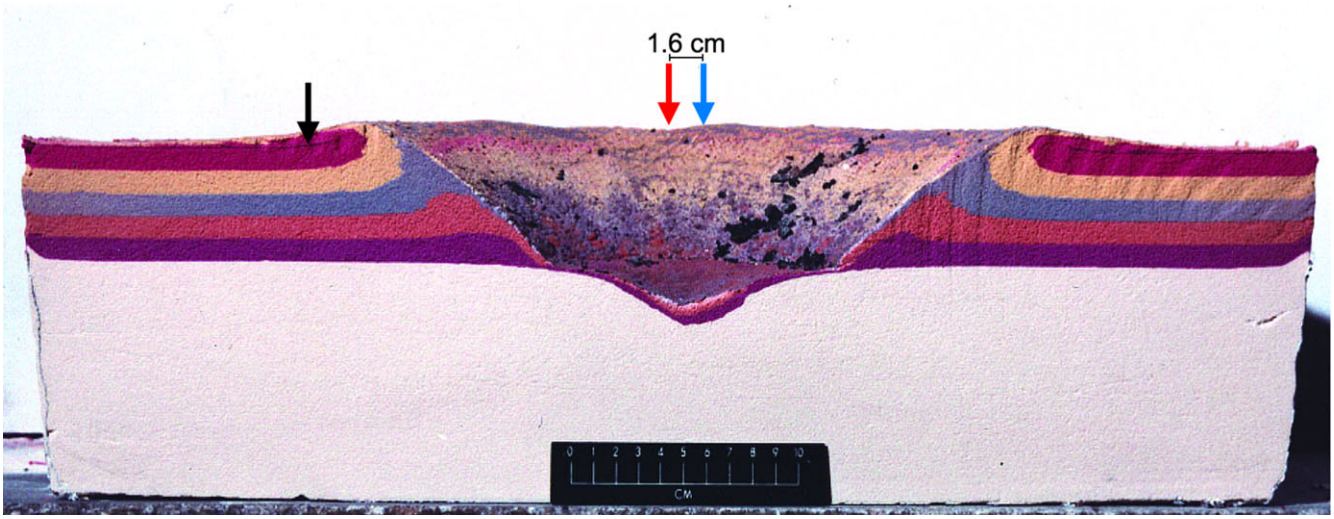


Fig. 4. Cratering experiment No. 420 (Tables 1a–c, Type 2): cross section of target after impact showing thin layer of hard black lacquer (black arrow) indicating the preimpact surface; the layer is cut off by the overturned flap at the crater rim; red arrow: center of target pan, blue arrow: center of crater; black particles inside the crater and veneer of white dust represent melt particles and shock-comminuted sand, respectively; scale at bottom in cm; diameter of the top of the complete cross section is 58 cm.

Table 2a. Target configurations used for Series I experiments with LEXAN projectiles.

Experiment No.	Target type	Vertical sequence of layers (from top to bottom)
387	Five colored layers, each 9 mm thick	Red-yellow-blue-orange-violet-plain
390	Plain sand	—
392	Plain sand	—
416	Plain sand	—
417	Plain sand	—
418	One colored layer (11 mm), four colored layers (9 mm)	Red-yellow-blue-orange-violet-plain
421	7 mm top layer with colored torus rings	Lateral sequence from center to rim: red-yellow-blue-orange
420	Five colored layers (9 mm) + epoxy powder	Red-yellow-blue-orange-violet-plain

per unit area on the ground plane of 11%. In cross sections of the crater of experiment No. 420, we found that the center of the crater was shifted away from the center of the target bucket by 1.6 cm (average of both sides of the cross section) (Fig. 4).

Also, some propellant gas (compressed hydrogen) enters the target chamber and may interfere with the ejecta cloud. This may explain why we observed that in the high-speed movies taken with a Nova camera at 7000 frames per second some asymmetry of the ejecta curtain, which would also explain the observed asymmetric mass distribution of the collected ejecta (see Fig. 8 and the Mass Distribution of the Ejecta as a Function Distance section).

METHODS OF NUMERICAL MODELING

In this work we use the iSALE-2D shock physics code (Wünnemann et al. 2006), which is based on the SALE (impact Simplified Arbitrary Lagrangian Eulerian) hydrocode solution algorithm (Amsden et al. 1980). To simulate hypervelocity impact processes in solid materials SALE was modified to include an elastoplastic constitutive model, fragmentation models, various equations of state (EoS), and multiple materials (Melosh et al. 1992; Ivanov et al. 1997). More recent improvements include a modified strength model (Collins et al. 2004) and a porosity compaction model (Wünnemann et al. 2006; Collins et al. 2011). The code has been successfully validated against laboratory cratering experiments and benchmarked against other well-established hydrocodes (Pierazzo et al. 2008; Davison et al. 2011; Güldemeister et al. 2015). For further details, see also http://www.isale-code.de.

Rheology Model

To model crater formation in sand we used a Drucker-Prager strength model that represents an appropriate rheological description of the behavior of

Table 2b. Target configurations used for Series II experiments with spherical aluminum projectiles.

Experiment No.	Depth of colored layer (cm)	Range of differently colored torus rings (cm)					
	0	1.5	3.5	5.5	8.5	11.5	14.5
529	0–0.9	Black	Red	Purple	Green	Blue	Orange
530	0.9–1.8	Black	Red	Purple	Green	Blue	Orange
531	1.8–2.7	Black	Red	Purple	Green	Blue	Orange
532	2.7–3.6	Black	Red	Purple	Green		
549	0–0.9	Black	Red	Purple	Green	Blue	Orange
548	0.9–1.8	Black	Red	Purple	Green	Blue	Orange
550	1.8–2.7	Black	Red	Purple	Green	Blue	Orange
534	1.8–2.7	Black	Red	Purple	Green		
533	2.7–3.6	Black	Red	Purple	Green		
Volume and mass of differently colored torus rings, (cm^3)/(g)		6.36/9.54	28.28/42.41	66.75/100.13	137.53/206.30	236.40/354.59	358.07/537.10

Table 3a. Details of the sampling plans of cratering experiments of Series I (see Tables 1a–c): (top) Radial ranges and areas of sampled rings on reference ground plane (total area corresponds to a 45° sector with a radial range of 18–105 cm), (bottom) angle of vertical catchers relative to ground plane; see also Fig. 2.

Sampled ring No.	Radial range (cm)	Sampling area (cm^2)
1	16–18	Not sampled
2	18–20	59.7
3	20–22	66.0
4	22–24	72.3
5	24–27	120.2
6	27–30	134.3
7	30–35	255.3
8	35–45	628.3
9	45–60	1237
10	60–75	1590.4
11	75–90	1943.9
12	90–105	2297.3
Total		8404.7
Sampled ring No.	Angle range (a) (°)	Area of quadrant (cm^2)
13	0–21.8	7602.1
14	21.8–32.9	4693.8
15	32.9–41.2	4343.1
16	41.2–46.9	3671.9
17	46.9–51.5	3583.7

granular material such as sand. In the model the shear strength Y is a function of the confining pressure P: $Y = Y_{coh} + \mu P$, where Y_{coh} is the cohesion and μ the coefficient of friction. At the time when the experiments were conducted neither Y_{coh} nor μ were determined for the given sand. It is also unclear whether the color coating of the sand grains affected the strength properties; for instance it is possible that the color increased the resistance against deformation at zero pressure (Y_{coh}) and/or reduced μ. The crater profile shown in Figs. 4 and 6 suggests that the bulk mechanical properties of plain and dyed sand may not have been identical. The nested crater structure similar to the morphometry of some small natural craters has often been interpreted as a result of different lithological layers with a denser and/or stronger half-space below a softer and/or less dense layer. However, in the example in Fig. 4, a small amount of epoxy powder was mixed into the sand, which may have also influenced the rheology of the sand. The crater profiles in most other experiments do not show a nested crater morphology and we consider the effect of the coloring to be negligible.

As no measurements of μ and Y_{coh} were available, we choose reasonable estimates from the literature (e.g., Wünnemann et al. 2011; Ormö et al. 2015) and varied the parameters slightly until we achieved a satisfactory match with the observed crater dimensions in the laboratory experiments of Series II. The same parameters were then also used for the models of Series I. We did not distinguish between plain and dyed sand in the models. All model parameters are listed in Table 4.

Crushing of Pore Space and Compression of Grains

Next to rheology, the shock compression of sand grains and the compaction of pore space significantly affect crater formation and the ejection process. We used the ANEOS (Thompson and Lauson 1972) for quartzite (Melosh 2007) to model the compression of sand grains and the ε-α-model (Wünnemann et al. 2006; Collins et al. 2011) to account for the crushing of pore space as a result of shock loading. A so-called crushing curve of the given sand to quantify the compaction behavior was also not available. The three parameters required for the ε-α-model were chosen according to

Table 3b. Details of the sampling plans of cratering experiments of Series II: (top) Radial ranges and areas of sampled rings on reference ground plane, (bottom) angles of vertical catchers relative to ground plane; see Fig. 2.

Sampled ring No.	Radial range (cm)	Area of quadrant (cm^2)
1	16–18	53.4
2	18–20	59.7
3	20–22	66.0
4	22–24	72.3
5	24–26	78.5
6	26–28	84.8
7	28–30	91.1
8	30–37.5	397.6
9	37.5–45	486.0
10	45–52.5	574.3
11	52.5–60	662.7
12	60–75	1590.4
13	75–90	1943.9
14	90–105	2297.3
Sampled ring No.	Angle range (°)	Area of quadrant (cm^2)
15	0–21.8	7602.1
16	21.8–32.9	4693.8
17	32.9–41.2	4343.1
18	41.2–46.9	3671.9
19	46.9–51.5	3583.7

previous modeling studies (Wünnemann et al. 2006; Güldemeister et al. 2013) and were then varied in a reasonable range for unconsolidated sand to match the observational crater dimensions in the experiments (Table 4). The initial porosity of the sand is given by $\phi_0 = 1 - (\rho_s/\rho_q)$, where ρ_s is the measured bulk density of the sand ranging between 1380–1558 kg m^{-3} for colored and plain sand, respectively, and ρ_q is density of the quartz grains with $\rho_q = 2670$ kg m^{-3}. Note, the bulk density of the colored sand is slightly lower than the density of plain sand. However, the density difference was found to be insignificant in some test models where a layer of the lower density, colored sand was included.

Aside from shock-induced compaction and crushing of pores granular material also experiences bulking due to shear. The effect of dilatancy (shear-bulking) has been investigated by numerical modeling in Collins (2014). We cannot rule out that dilatancy may change the crater morphometry in our models slightly; however, as we focus on the ejection process, we consider this effect to be negligible.

Modeling Experiments

The main difference between the individual experiments in Series I and II, respectively, was the location of the colored sand layers. Therefore, it was not necessary to model each individual experiment as the ejecta distribution in the experiments can be only interpreted as a whole of all experiments in Series I (Nos. 387, 416–418, 420–421) and II (Nos. 529–532 and Nos. 533–550). Variations in impact velocity and possible slight differences in the constitutive behavior of the colored sand were neglected in the models. We chose experiment No. 529 (Series II) and No. 418 (Series I) as representative and suitable reference for the comparison of crater morphometry (see data in Tables 1a–c).

As we were primarily interested in modeling the ejection process, all simulations were stopped at a time when the approximate transient crater was reached to reduce the computational costs. Simulating the entire crater formation process including slumping of a relatively small fraction of sand along the over-steepened crater wall and the accumulation of material at the bottom of the cavity is computationally very expensive (see discussion in Pierazzo et al. 2008). The size of the transient crater D_t in the experiments could not be recorded and the actual diameter for comparison with models was only estimated. As mentioned above we consider a ratio D_a/D_t of 1.08 (see the Quantitative Techniques for the Analyses of Craters and their Ejecta section).

In all models of crater formation the projectile was resolved by 10 cells per projectile radius (CPPR) and was composed of aluminum (Series II) and LEXAN (Series I). This corresponds to a space increment of 0.395 mm for Series I and 0.318 mm for Series II. Thus, the thinnest, 9 mm thick, colored layers were resolved by 28 cells. The total grid consisted of 1000 × 900 cells in the high-resolution zone. Beyond this zone the grid size was gradually extended by a factor of 5% over 50 cells in each direction to ensure that the boundaries of the computational domain did not affect the model results. A few models were also carried with a resolution of 60 CPPR to resolve the zone of high shock pressures, adequately. These models were stopped when the shock pressure had decayed below the level where any shock or plastic wave-induced modifications occur (~10 MPa).

Determination of Ejection Parameters

Continuum models such as iSALE describe the deformation and displacement of material in the target relatively well; however, the ejecta curtain and the ballistic transportation of material is generally only poorly resolved (a few computational cells) in such models. Tracers are the most appropriate method to record the trajectory of material during the course of

Table 3c. Details of the sampling plans of cratering experiments of Series II (see Table 1a–c): Sampling quadrants and ranges (for numbers 1–19 see Table 3a) and types of measured data; note that shocked particles were also sampled from inside the craters; note that not all quadrants were completely sampled within rings 1–19.

Experiment No.	Mass and color determination Quadrants				Analysis of grain size and shocked particles Quadrants			
	A	B	C	D	A	B	C	D
529	2–19	2–19	2–19	15–18			1–19	
530	15–18	1–18	1–19	15–18			1–19	
531	2–7, 15–19	15–19	1–19	8–18			1–19	
532	1–19	15–19	15–19	1–18	1–19			
549	1–19	1–19	15–19	15–18	1–19			
548	1–19	1–19	15–19	15–18		2–19		
550	2–19	2–19	15–19	15–18	2–19			
534	1–7, 15–19	1–19	1–19	1–7, 15–18		1–19		
533	2–7, 15–19	2–19	2–19	2–7, 2–18		2–19		

crater formation. We placed massless tracers in each computational cell representing the matter that was originally located in the cell throughout the simulation. When a tracer moves above the preimpact target surface it is considered to be ejecta and we recorded the angle β and velocity v of ejection. Figure 5 shows the normalized ejection velocity (v/v_i, where v_i is the impact velocity of 6.44 km s^{-1}) and angle (β) as a function of normalized launch distance (r/a, where a is the projectile radius of 0.318 cm and r is launch distance) for experiment No. 529 (Series I). Following Housen and Holsapple (2011) we define a power-law regime (between dashed vertical lines in Fig. 5). Early ejecta partially representing jetted masses (at a radial distance <6 projectile radii a) are not resolved reasonable well in our simulations (Johnson et al. 2014). The velocity of late ejecta (at a radial distance >35a) are strongly affected by gravity and strength and our tracer-approach breaks down at this distance. Generally, the ejection parameters determined in our models are in good agreement with previous modeling studies using the same approach (Collins and Wünnemann 2007; Zhu et al. 2015) and measured ejection velocities and angles from PIV (e.g., Cintala et al. 1999; Hermalyn and Schultz 2010). Material is ejected initially at ~45°. With increasing distance the ejection angles vary bounded by an envelope of ~30°–45°. The ejection velocity decreases with distance proportional to a power-law with an exponent of −2.3, which is in good agreement with the value of 2.4 (=−1/0.43 = 1/μ) for dry soil according to Housen and Holsapple (2011).

The mass of the ejecta is given by the amount of matter in the cell where the tracer was originally located. Provided that the flight of ejecta is ballistic, it is possible to reconstruct the parabolic trajectory of each tracer (and its "mass"), the velocity, and distance where it hits the ground or vertical catchers. Integration over all ejected tracers allows for modeling the distribution of ejecta, the mass, and the place of origin as a function of radial range for comparison with the integrated results from the experimental craters with annuli of color-coded sand at different depths. Note that there are differences in resolution comparing ejecta distribution from models and experiments (1) in models the resolution for the place of origin is given by the cell size (0.3–0.4 mm) and the mass of tracers is summed up across intervals of 20 mm radial distance; (2) in the experiments the resolution for the place of origin of grains is given by the width of the rings (20–30 mm), the thickness of the layers (9 mm), and the width (opening angle) of the vertical catchers varying between 20 and 30 mm.

RESULTS

Crater Morphometry

Although the focus of this paper lies on the ejection process, any model must also reproduce the observed crater morphometry in the experiments satisfactorily. Figure 6b shows all crater profiles with two perpendicular profiles for each experiment of Series II. The upper frame (Fig. 6a) depicts a sketch of the mechanical profilometer used to determine the morphometry of the experimental craters. For any given data set, there is some difficulty in comparing modeled crater shapes and sizes against the experiments because the models were only carried out until the transient crater was reached. Despite this deficiency we plotted all crater profiles of Series II (No. 529–550); spherical aluminum projectile) and one crater profile of Series I (No. 418; cylindrical LEXAN projectile) on top of

Table 4. Settings and material parameters used in the iSALE models.

Settings		Series I		Series II
Resolution in cells per projectile radius (CPPR)				
Crater formation model (see Table 1a)		10		10
Shock wave propagation model (see Table 1b)		60		60
Modeled experiment No. (used as reference)		418		529
Projectile geometry		Cylinder		Sphere
High–resolution grid size (crater formation model, r × v in cells)		1000 × 900		1000 × 900
Strength model parameter	Variable	Sand	Aluminum projectile	LEXAN projectile
Strength model		Drucker-Prager	Hydrodynamic	Hydrodynamic
Poisson rate	γ	0.30	0.5	0.5
Yield strength at zero pressure (MPa)	Y_0	0.00	–	–
Yield strength at infinite pressure (MPa)	Y_{inf}	100	–	–
Friction coefficient	F	0.56	–	–
Porosity model parameters (ε-α-model)[a]				
Initial porosity	ϕ_0	0.42	–	–
Distension at onset of crushing	α_e	1.71	–	–
Distension at transition to power-law	α_x	1.29	–	–
Exponential compact rate		0.99	–	–
Ratio of porous/nonporous sound speed	χ	0.33	–	–
Equation of state parameters		ANEOS[b] quarzite	Tillotson aluminum	Simplified Tillotson[c]
Density of matrix (kg m^{-3})	ρ	2670	2700	1216
Tillotson parameter A (bulk modulus at zero pressure, GPa)	A	–	75.2	9.94
Tillotson parameter B (constant, GPa)	B	–	65	–
Tillotson parameter E_0 (constant, MJ kg^{-1})	E_0	–	5	–
Tillotson parameter a (dimensionless constant)	a	–	0.5	0.6
Tillotson parameter b (dimensionless constant)	b	–	0.5	–
Tillotson parameter α (dimensionless constant)	α	–	5	–
Tillotson parameter β (dimensionless constant)	β	–	5	–
Tillotson parameter Eiv (specific internal energy of incipient vaporization, MJ kg^{-1})		–	3	–
Tillotson parameter Ecv (Specific internal energy of complete vaporization, MJ kg^{-1})		–	13.9	–

[a]Parameter definition according to Collins et al. (2011).
[b]ANEOS parameters for quartzite at given at Melosh (2007).
[c]Tillotson parameter approximate a linear liquid EoS, Melosh (1989).

snapshots of the approximate transient crater from numerical modeling in Figs. 6c and 6d. The suite of profiles illustrates the variations of crater morphometry in the experiments and deviations from perfect symmetry. The apparent crater diameter D_a at the reference surface ranges between 26.30 cm and 28.35 cm and the depth between 5.3 cm and 7.4 cm, respectively (Table 1b). The variations primarily result from differences in impact velocities (5.86–6.46 km s^{-1}); however, it is important to note that the experiment with the largest amount of kinetic energy (No. 530, Table 1b) did not generate the largest crater in terms of volume (No. 533, Table 1b). A possible explanation for the fact that crater size does not exactly scale with impact energy may be that the location of the colored layers in the target had some effect on crater formation due to the above mentioned small differences in terms of the material properties of colored sand in comparison to the uncolored sand (density, coefficient of friction, cohesion). As the crater diameter D_a varies only by about 7%, we consider this effect to be small, thus justifying the general assumption that the outcome of all experiments in terms of crater formation and ejecta deposition is essentially the same.

First, we chose experiment No. 529 (red line, Fig. 6c) as reference for comparison with the modeled event. The blue line in Fig. 6c shows the numerical model at the point when the transient crater was fully developed and the simulation was stopped. The kink in the ejecta curtain indicates that the excavation flow has ceased and that any additional growth in crater diameter only occurs due to gravity driven collapse.

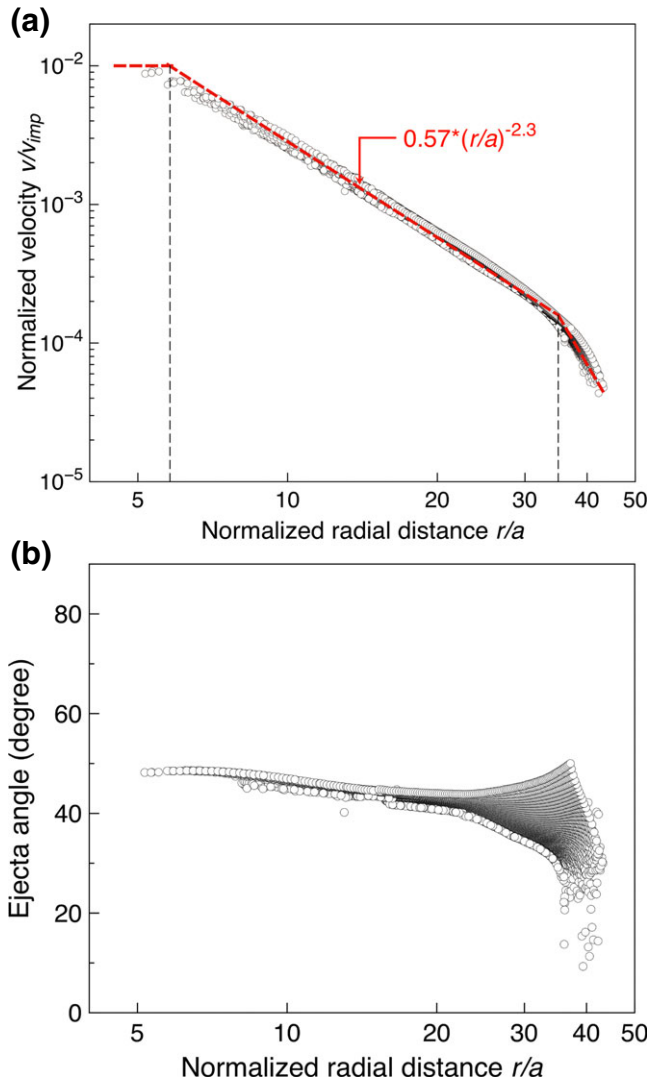


Fig. 5. Normalized ejection velocity (a) and angle (b) as a function of normalized launch distance (r/a, where r is radial distance and a is the projectile diameter) determined from model of experiment No. 529 (Series II). Solid line (a) represents a power-law fit bounded by dashed vertical lines.

The diameter of the transient cavity in the numerical model is D_t = 24.6 cm. According to the above estimate of D_a = 1.08 D_t, we obtained an estimated final crater diameter in the models of D_a = 26.6. This value deviates from the measured final apparent crater D_a = 26.85 cm in experiment No. 529 only by 1%. The measured crater depth in the experiment (d = 5.8 cm, No. 529) is inevitably shallower than the transient crater depth in the model (d_t = 6.5 cm) due to the accumulation of slumped material at the bottom of the crater.

For Series I (cylindrical LEXAN projectile, No. 418), the modeled transient crater diameter D_t is 30.58 cm, which corresponds to an estimated final apparent crater diameter of D_a = 33.0 cm (see Fig. 6d). The measured value of the crater diameter is D_a = 27.15 cm which is 18% smaller than in the model. The observed crater depth is d = 5.8 cm; the transient crater depth in the model is d_t = 6.5 cm.

We can only speculate on the reasons for the increased divergence in comparison to the very good agreement between models and experiments for Series II. One reason could be that the material models for the LEXAN projectile is only a very crude approximation. The equation of state model we used describes the behavior of a liquid that undergoes little compression so that the pressure can be estimated by a linear function of density (see Melosh 1989; AII.1, p. 230). In this model the only material parameters are the initial density and the speed of sound (or bulk modulus; see Table 4). Although this simple EoS has given reasonably good results in models of impact experiments in the range of some 100 km s^{-1} (e.g., Ormö et al. 2015) it may oversimplify the conditions at the velocities at an average of 6.6 km s^{-1} (Series I) in this study. We also assume the impactor to be strengthless (hydrodynamic behavior) for both the LEXAN and the aluminum projectile. As a consequence in our models, the projectile distorts and expands laterally ("pancaking") upon penetration of the target forming a thin veneer lining the crater floor that is not resolved sufficiently in our models (initially, the projectile is only resolved by 10 CPPR). This simplified material behavior does not account for disruption of the projectile. In the case of the aluminum projectile the material response is more ductile and our model may pose a better approximation than for the LEXAN projectile that is probably more brittle and undergoes more extensive fragmentation. More detailed studies of the projectile behavior during the penetration and disruption are required to assess how the constitutive model for the projectile affects the crater formation process.

Another reason for the increased divergence between model and experiment in Series I could be the cylindrical geometry of the projectile. In our model we assume that the projectile hits the target with the flat surface. In the experiment the impactor strikes the target in some arbitrary orientation. However, due to the large crater efficiency (the crater is much bigger than the projectile) we consider the latter point to be less important to explain the increased divergence between model and experiment for Series I.

Mass Distribution of the Ejecta as a Function Distance

Figure 7 shows the comparison of the mass of the deposited ejecta per unit area versus radial distance

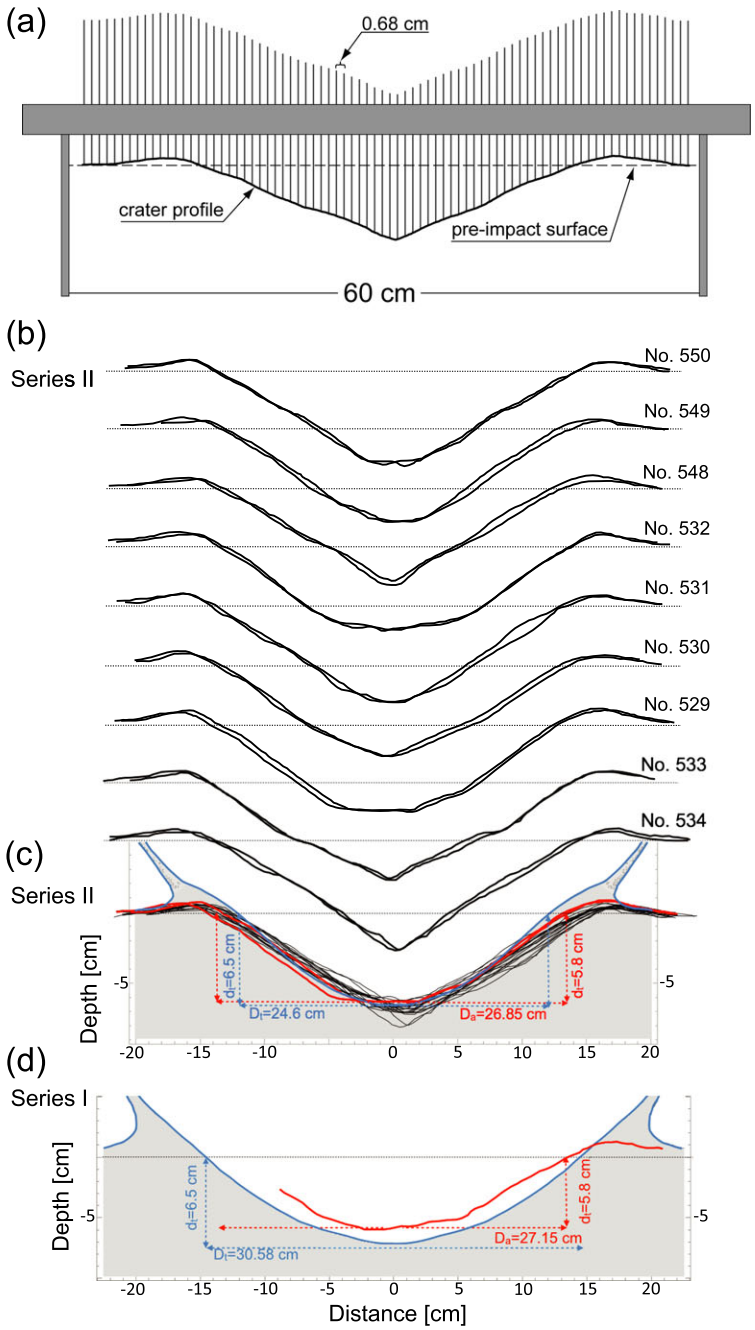


Fig. 6. Crater profiles of experiments in comparison to numerical models. a) Sketch of mechanical profilometer to measure crater morphometry (80 vertically adjustable rods with a spacing of 0.68 cm); b) crater profiles (two perpendicular profiles for each experiment) of Series II; c) comparison of crater profiles (red highlights experiment No. 529) with transient crater shape from numerical model of experiment No. 529 (blue line); d) comparison of crater profile from experiment (red) with transient crater from numerical model (blue) of No. 418.

between the different experiments of Series II (a) and Series I (b) and the numerical models of the reference experiments No. 529 and No. 418. For reasons explained above (slight variation in impact velocities, offset between the point of impact and the geometric center of the ejecta catcher) the mean standard deviation of the ejected mass for the different experiments at different distances is 14% (Series II) and 21% (Series I), respectively. We fitted a power-law to the data excluding distances >90 cm. At this distance the mass of the deposited ejecta slightly increases in some experiments due to particles that were not captured by the vertical

catchers and bounced back from the catchers at the wall of the target chamber. In fact, no ejecta were recorded at angles higher than 51° relative to the ground plan using the device shown in Fig. 1 and the mass in the uppermost (5th) vertical catchers is only 1% of the mass found in the catcher below (4th) or 0.03% of the mass found at the crater rim.

The mass of the ejecta decreases with distance according to a power-law with an exponent of −3.16 (Series II) and −3.27 (Series I), respectively. Generally, the results of Series I and II are consistent and also in good agreement with data from Hörz et al. (1983) who observed a decrease of the ejecta thickness (Bunte Breccia) at the Ries crater according to a power-law with an exponent of −3. The modeled ejecta mass as a function of radial range is very similar to the ejecta distribution in the experiments. The power-law fit to the models of the reference experiments No. 529 and No. 418 yields a decay exponent of −3.28 and −3.41 for Series II and I, respectively, which is very close to the experimental values. The modeled ejecta distribution does not show any evidence for two regimes of decreasing ejecta thickness (steeper decay in the vicinity of the rim than at further distance) as suggested by Stöffler et al. (1975); however, the tracer method in the models to determine the ejecta distribution does not account for structural uplift at the crater rim, which may be the reason for a steeper decay of the ejecta thickness in the vicinity of the crater.

It should be noted that for larger distances the data quality decreases in the experiments due to the relatively small amount of sand grains that were deposited as distal ejecta. The same holds true for the numerical models where the number of tracer particles is even smaller than the number of grains in the experiments.

Relationship Between Preimpact Location and Ejection Distance of Particles

An important goal of this study is to relate the distance of ejected sand grains and their degree of shock modification with their initial location in the target. This question has been tackled experimentally in Stöffler et al. (1975) by colored layers allowing for the determination of the preimpact depth of grains underneath the target (1-D reconstruction, Series I), but not their radial distance to the point of impact. In this study (Series II) the colored sand grains were initially arranged in concentric rings at varying depth (see Fig. 3), which enables a 2-D (depth and radial distance) reconstruction of the place of origin for ejected masses.

First, we focus on the reference experiment No. 529 (Series II) where the rings of colored sand are located in the uppermost layer and where almost all material is excavated and ejected, except a small region right underneath the point of impact. Figure 8 shows the ejected mass per unit area as a function of radial range to the point of impact. The initial configuration of the colored sands was described in Fig. 3 and is superposed on the crater profile in the bottom frame of Fig. 8. The two opposite sectors of catchers (A) and (C) (Fig. 2, Table 3b) show approximately the same decrease of the total mass with distance, but the proportion of different-colored sand differs for catchers of equal radial distance to the bucket center. The obvious reason for this asymmetric distribution is given by the offset between the point of impact and the geometric center of the color-coded rings, the nominal aim point, as discussed above.

How much the point of impact is shifted out of the geometric center in the experiments is not exactly known. It can be reconstructed only for experiment No. 420 by measuring the offset between the deepest point of the crater and the geometric center, which is $\Delta r \simeq 1.6$ cm (see Figs. 3 and 8 and the Quantitative Techniques for the Analyses of Craters and their Ejecta section). Despite these asymmetries the ejecta close to the crater rim (<22 cm in sector A and <37.5 cm in sector C) mainly originate from the outermost ring (orange). At larger distances the ejecta are dominated by material from the blue ring. Interestingly, almost at all distances black grains apparently from the inner most (black) ring are present (this may be due to a technical problem as discussed at the end of this section). As we assume that the projectile hit the target with some offset toward sector C (lower frame in Fig. 8) most material at radial ranges of 75–90 cm is from the outermost green and blue rings (upper left frame in Fig. 8) while in sector A (upper right frame in Fig. 8) mostly black and red grains from the innermost rings were found. Samples from larger distances were presumably contaminated by particles that bounced back from the vertical catchers. In addition, colorless grains dominate at $r = 60$–90 cm in sector A in contrast to sector C. Note in the logarithmic representation of Fig. 8 the fraction of black, red, violet, and green grains are overemphasized. The absolute amount of these grains is small relative to the number of blue grains.

We assume that the asymmetric distribution of the ejecta is a consequence of the offset between the point of impact and the geometric center of the colored annuli. Modeling of the asymmetric distribution of ejecta on a 2-D cylindrically symmetric grid is not possible. However, the asymmetric conditions in the experiment (offset Δr) can be approximated by adjusting the radial distance of the colored rings from the point

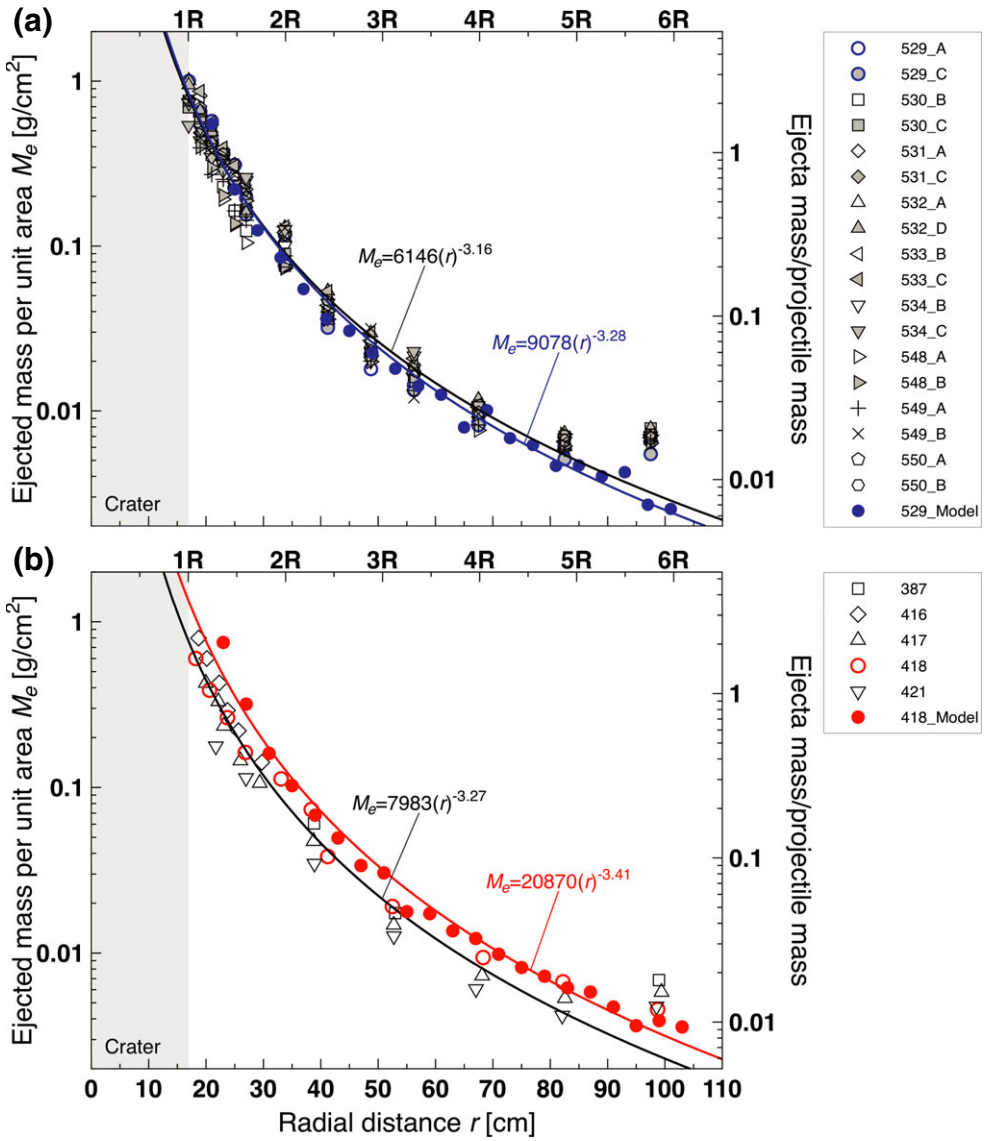


Fig. 7. Ejected mass per unit area Me (right: normalized mass of projectile) as a function of radial distance r (top: normalized by crater radii R) from the point of impact for all experiments of Series II (a) and 5 of Series I (b) in comparison to results from numerical modeling of experiment No. 529 (blue dots) and experiment No. 418 (red dots). The blue and the red line correspond to a power-law fit to the modeled data. The black lines correspond to a power-law fit to the data from the experiments. The gray shaded area indicates the approximate size of the crater (note, the actual final crater radius varied in the experiments from 15.9 to 16.9 cm in Series II and from 14.8 to 16.4 cm in Series I, see Tables 1a–c). The small increase in mass for ejecta at distances >90 cm is due to material that was not captured by vertical catchers, bounced back, and fall into the catchers on the ground.

of impact until the best agreement between models and experiments is produced. Note, by this method the actual distribution can only be approximated as the real conditions pose a 3-D problem. Figure 9 shows a comparison between the ejecta distribution obtained from the model of experiment No. 529 with Δr = 1.5 cm and the observed ejecta distribution in sector A. Close to the crater rim the ejecta distribution is very similar and is mainly composed of orange tracers/sand grains. Up to a distance of 65 cm (model) and 55 cm (experiment), respectively, the ejecta are dominated by blue tracers/grains. Further away from the impact site, only tracers that originate from the green ring or from larger depth (white tracers, >9 mm depth) are deposited according to the model (the very small fraction of violet tracers at 65 cm is negligible). In contrast, only a relatively small fraction of green grains was observed in the experiments and mainly uncolored grains and material from the innermost annuli (black, red, violet) are present at larger distances (>55 cm).

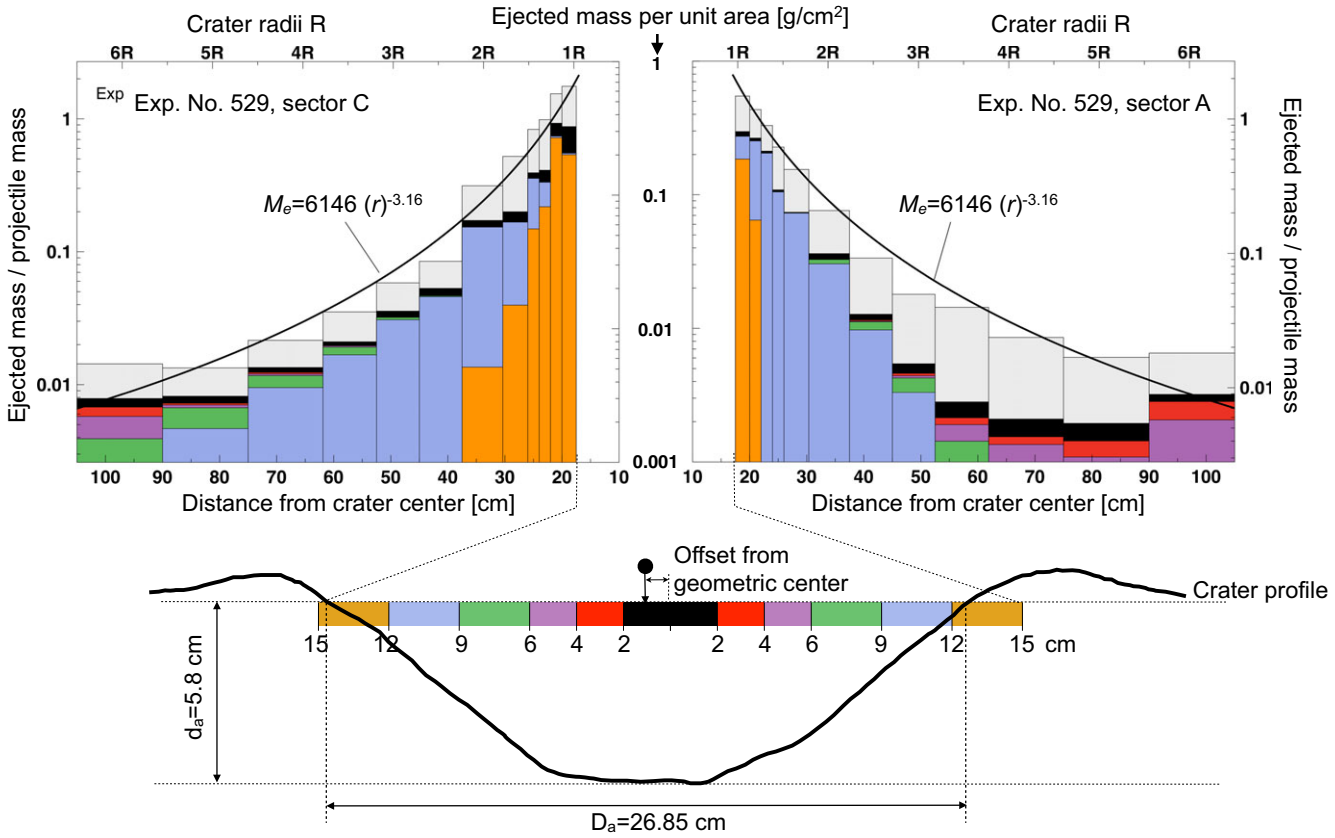


Fig. 8. Distribution of ejected mass as a function of distance in experiment No. 529 (Series II) for two opposing catcher sectors A (right side) and C (left side). The original location of colored sand grains arranged in centric rings and the crater profile are shown in the lower frame. Note also the offset between point of impact and the geometric center of the rings. The bar-chart underneath the power-law decay function (black lines, according to Fig. 7) indicate the fraction of colored grains at a given distance. Gray color represents the fraction of grains with no color (plain sand) that derive from depth deeper than 9 mm.

Integrating the distribution of color-coded ejecta from all Series II experiments using buried marker annuli, one can reconstruct specific ejection zones. The bold black line in Fig. 9 (lower right frame) marks the zone from where material is ejected beyond the rim of the transient crater. This is generally referred to as the excavation cavity. Below this line sand grains are displaced from their original location but not thrown out of the crater. The two thinner black lines mark zones from where material is ejected to >2 and >7 crater radii, respectively. The dashed black lines indicate the original interpretation (e.g., Stöffler et al. 1980b). Due to some technical issues that we discuss at the end of the section, the course of these latter lines is questionable. The white dashed line represents the expected trend of the excavation zone.

The numerical models enable a more detailed reconstruction of the zones of ejection as each tracer can be easily tracked back to its original target location. We distinguish between seven zones in Fig. 9 (lower left frame). The contours indicate the ejection distance between >1 and >7 crater radii R and can be directly compared with the experiments. The radial extent of the zones is very similar in model and experiment, but the excavation depth is deeper in the models. However, as suggested by the models, the experiments may not provide reliable data for the deepest excavations as discussed below.

The ejecta distribution and the inferred excavation zones of the experiments agree with the models only in part and there are some discrepancies that require explanation. The most striking mismatches between model and experiment are: (1) The lack of green grains in the ejecta at distances >60 cm and (2) the presence of black grains, from the central location, at all distances; all of this material should be ejected beyond the horizontal catchers (>100 cm). These deviations between experiments and model may be explained as follows.

1. It cannot be ruled out that some grains lost their color during the ejection process and were mistaken for colorless instead of green. Note, due to the irregular shape of the sand grains they were not

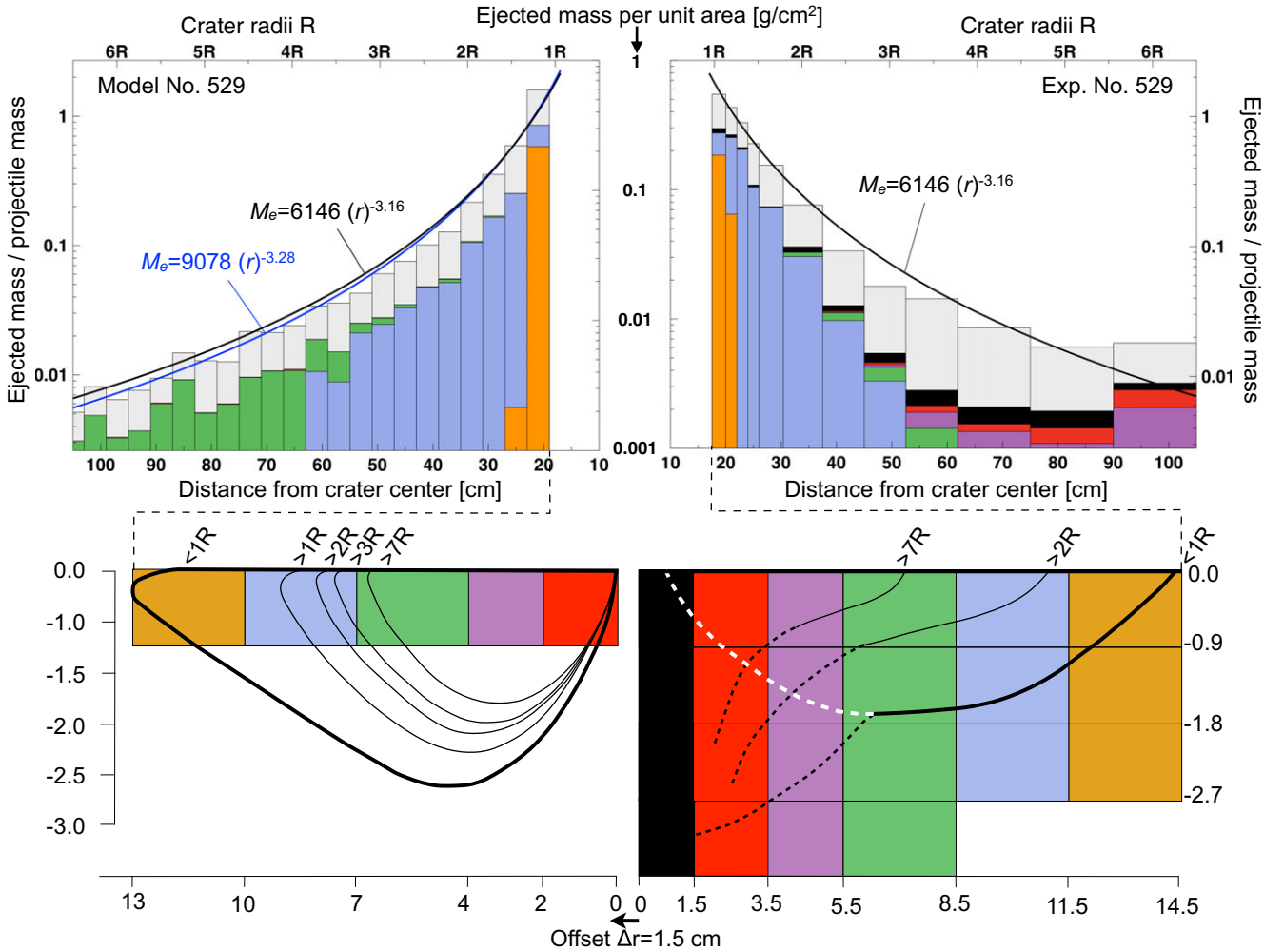


Fig. 9. Comparison of ejecta distribution in experiment (No. 529, sector A, colored rings in top layer) and model (upper frame). In the model it was assumed that the point of impact was shifted by 1.5 cm away from the geometric center of the color rings. The original location of colored rings is shown in the lower frames. Based on the analysis of the distribution of colored sand in all experiments of Series II in two opposite catcher sectors three zones were reconstructed (lower right frame): (I) excavation zone (<1R), (II) zone of ejection >2 crater radii, (III) zone of ejection >7 crater radii (after Stöffler et al. 1980a, 1980b, 1983); the dashed lines indicate the original interpretation, which has been revised (see text); the white dashed line corresponds to the expected trend of the excavation zone. In the model (lower left frame), we distinguish between seven zones of ejection distances.

entirely coated with dye. This would explain the deficiency of green markers in the ejecta and the large fraction of colorless grains in the experiments, while the total amount of all ejecta is similar in model and experiment. Sand grains from annuli with other colors were probably less affected as the grains were either ejected early (black, red, and violet) or experienced less mechanical loading toward the end of the excavation phase. The trajectory of the green sand grains below the target surface and before they are ejected is approximately the longest in comparison to grains from other annuli. Along this path the grains probably experienced a lot of drag that may explain why a relatively large number of green sand grains may have lost their color. The lack of green sand grains in the ejecta distribution is the main reason for the estimated shallow excavation in the experiment, which is in strong contrast to the general understanding of crater mechanics.

2. It is also possible that the black lacquer that was sprayed on the target to identify the preimpact surface blackened a layer that was multiple grains deep. This would explain why a small fraction of black grains was basically found at all distances, which disagrees with the tracer distribution of the model which ejected all black surface tracers to distances larger than observed in the experiment.

3. Although we tried to correct for the deviation of the impact site from the geometric center of the rings in our models, it is only an approximation to a 3-D problem. It is difficult, therefore, to estimate how the asymmetry in the experiments has affected the measured ejecta distribution.
4. Nonballistic effects including the interaction of the gas flow from the gun with ejected particles may have also influenced the observed ejecta distribution. Nonballistic particle movement can be observed in the high-speed movies of the ejection process in plan view of experiment No. 420, where some material (glowing shreds of melt) is clearly traveling vertically upwards. These particles are probably driven by vortical flows (vortexes) of the propellant gas coming out of the launch tube. Note that this observation was only made in this specific experiment, where the sand was admixed with epoxy powder allowing solidification by baking for subsequent cross cutting of the entire crater (Fig. 4). At this point, one can only speculate about the role of nonballistic processes. We consider the effect to be small, but apparently not negligible for the interpretation of the experimental results.

Whether other physical effects, such as atmospheric drag, play any additional role will be discussed below.

Types of Shocked Particles

Particles recovered from the ejecta catchers and from the crater floor were microscopically analyzed to investigate shock-induced modifications. This has not been done in detail before in any cratering experiment, much less as a function of source area and final position within the ejecta blanket. Detailed characterization of shocked particles and their modal abundance within our individual ejecta samples represents thus a unique data set that relates to the characterization of the decaying shock front within the excavation cavity. We observe three main types of shock metamorphic particles resulting from melting, aggregation, and comminution processes. Based on previous attempts to calibrate the shock effects observed in quartz sand and porous sandstone (Kieffer 1971; Kieffer et al. 1976; Stöffler and Grieve 2007; Kowitz et al. 2013), we assign shock pressure ranges to the main types of shocked particles in the following description of the particles. Note also that the peak shock pressures at the point of impact are according to the models ~23 GPa and ~59 GPa in Series I and Series II experiments, respectively. More details and a comparison with modeling results are given in the Shock Wave Attenuation and Zones of Shock Metamorphism section.

Melting

Melt particles occur inside the crater and throughout the entire ejecta blanket. They display a layered structure from the interaction of the shock decomposed plastic projectiles or of the shock molten metal projectiles with molten and shock lithified sand.

Melt particles produced with LEXAN projectiles have a black top coating with decomposition products of the polycarbonate (mainly carbon) underlain by highly vesiculated silica glass to which shock lithified sand is usually attached (Figs. 10a and 10b). These aggregates of irregular shape range in size from about 0.3 to 3 cm. Their bottom side contains clasts of shocked quartz grains with planar fractures and planar deformation features (pdf; Fig. 10c) grading into dense aggregates of quartz grains without pdf compacted to zero porosity (Fig. 10a). These composite particles record a total shock pressure range from ~23 GPa to roughly 1 GPa from top to bottom and their upper zone containing projectile melt and shock-fused quartz (lechatelierite) most probably formed at pressures larger than some ~5 GPa (Zones 1–3 in Fig. 12; see also Fig. 16 in the Shock Wave Attenuation and Zones of Shock Metamorphism section).

Melt particles produced with aluminum projectiles in Series II also have a black coating of molten aluminum attached to complex aggregates of shock-fused and shock-lithified sand with a distinct zoning of decreasing shock intensity (Figs. 11 and 12), which reflects a pressure range from about 59 GPa (peak pressure at the point of impact from models) to about 1 GPa. The upper melt-bearing section (zones 1–3) is presumably formed above ~5–13 GPa with partial melting starting at ~5 GPa (lower zone 3) and complete melting (lechatelierite) at ~13 GPa (upper zone 3). The shock effects in these zones are very similar to those described above for Series I experiments. In zone 4, the original dye color of the sand may be erased near to the melt-bearing zone 3 but may be preserved in the lower part depending on the preimpact position of the layer of colored sand. The complete zoning is schematically presented in Fig. 12 (Jammes et al. 1983; Jammes 1984) and will be described in more detail below (see also corresponding model calculations given in Fig. 16).

Aggregation

The second type of shock metamorphic particles lacks any coating with projectile material and any melt phases (Fig. 13). Like melt particles, they also occur inside the crater and throughout the complete ejecta blanket. They have sizes generally >0.2 mm ranging up to about 0.5 cm. We distinguish two types of such particles.

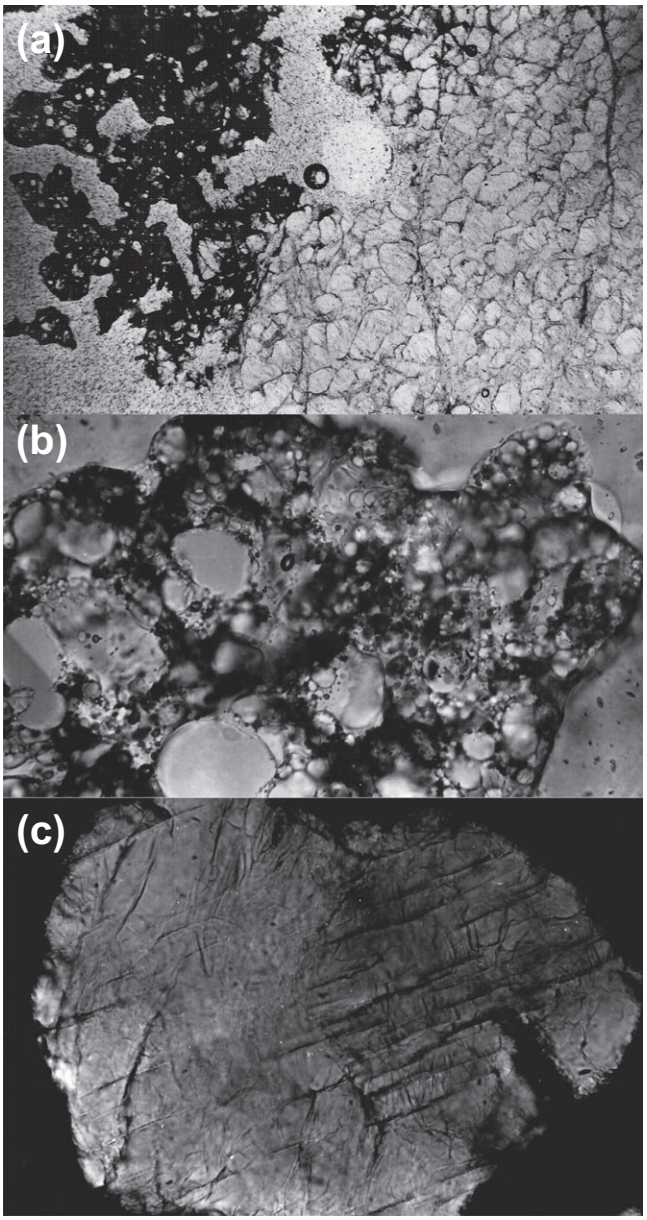


Fig. 10. Shock-fused and shock-lithified particles of experiment No. 387 of Series I with one particle (a) showing attached black melt crust (projectile + shock-fused quartz); (b) vesiculated silica melt with inclusions of shocked quartz; and (c) quartz grains with planar fractures and planar deformation features; photos taken with the polarizing microscope from thin sections in plane polarized light; widths of photos: 3 mm (top), 0.8 mm (middle) and 0.33 mm (bottom). See also Fig. 11 for comparison.

1. Shock lithified aggregates preserving the preimpact coloring of grains
2. Shock lithified aggregates without color originating from layers of plain sand

These shock lithified aggregates either retained the dye color of the nonfragmented quartz grains or they lack any color depending on their preimpact location. They may even preserve the boundary between two neighboring colors. They probably have experienced shock pressures ranging from about 1 GPa to about 5 GPa, which corresponds to the orange zone (0.9–5.5 GPa) in Fig. 16.

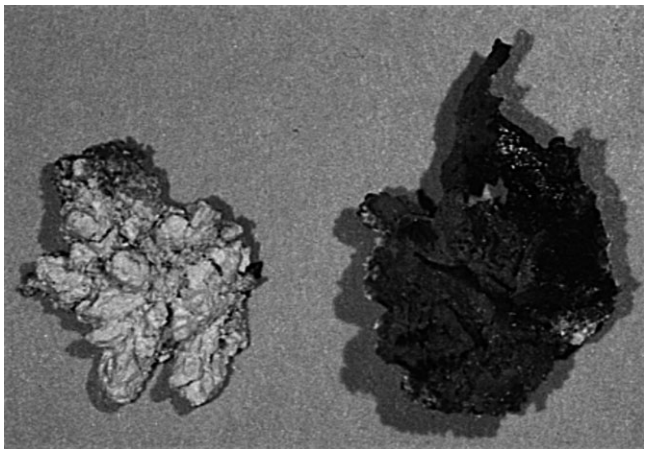


Fig. 11. Melt particle from experiment No. 530 of Series II with coating of molten aluminum on the top side (right) and shock lithified quartz sand aggregates at the bottom side (left); width of photo: 2.5 cm.

Comminution

Fragmented quartz grains represent the third type of shocked particles. Comminuted sand grains occur predominantly inside the craters and in the most distant ejecta, but are present throughout the ejecta blanket. As the vast majority of the original quartz grains covers the size range of 0.063 mm to 0.25 mm, the comminuted grains are mostly <0.125 mm in size and can be as small as a few micrometers (Fig. 13b). We assume that they form at shock pressures <~1 GPa (see the Mass Distribution of Shock Metamorphic Particles as a Function of Distance section and Fig. 16).

Reaction Zone Between Projectile and Target

The peak pressures at the impact point of Series I and II experiments are 23 GPa (experiment No. 418) and 59 GPa (experiment No. 529), respectively, according to the numerical models. The corresponding temperatures are high enough to decompose the plastic projectiles to carbon (Series I) and to melt the aluminum projectiles as well as an uppermost thin layer of sand as shown in Fig. 12. For Series II experiments, the thermal reactions of aluminum and quartz have been studied in some detail previously (Jammes et al. 1983; Jammes 1984), yet they were recently reanalyzed with high-resolution analytical tools (Hamann et al. 2016). Here, we present a short summary.

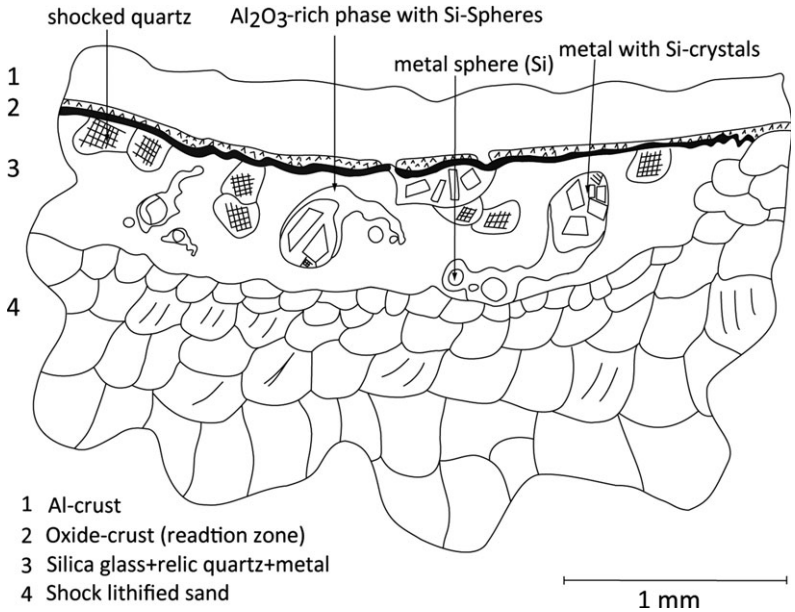


Fig. 12. Generalized cross section of a melt particle with top layer of molten Al-projectile and zones of decreasing shock metamorphism in the attached shock lithified sand aggregate; sketch based on observation from experiment No. 530; see text for more details; from Jammes et al. (1983); the lower part of zone 3 roughly corresponds to the green zone in Fig. 15 (5.5–13 GPa) grading into the zone of complete melting (13–59 GPa).

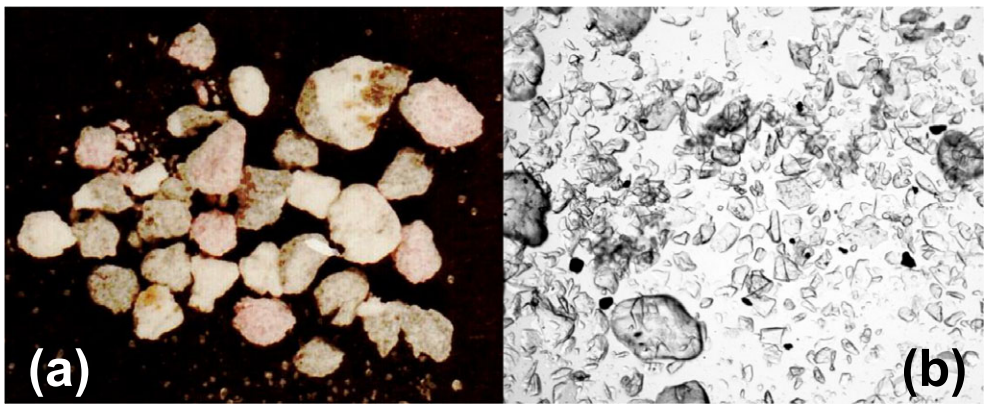


Fig. 13. a) Shock lithified sand aggregates of experiment No. 530; some aggregates still retain the color coding of the preimpact red layer; width of photo: 1.1 cm; b) shock-comminuted sand from the central bottom of crater (experiment No. 530) with grain sizes ranging from about 0.001 to 0.15 mm (nonfractured grains at left).

The aluminum projectiles melted upon impact and now form a thin crystalline crust on the topside of melt particles (Fig. 11; zone 1 of Fig 12). This process leads to a very distinct crystallization texture of this crust in which the trace elements of the projectile (Cu, Mn, Fe) form a thin network of eutectic phases surrounding large roundish bodies of Al that are strongly depleted in these elements (Figs. 14 and 15). The Al-crust is attached to a narrow reaction zone (Fig. 12, zone 2), which consists mainly of Al_2O_3 and finely dispersed metallic Si formed by an alumothermic reaction between Al and quartz. It forms a sharp contact to the underlying layer (zone 3) of vesiculated quartz glass (lechatelierite) in which relic grains of shocked quartz with planar deformation features as well as irregularly shaped and spherical metal particles are embedded (Fig. 14; see also Figs. 10c and 10d). The metal particles consist of an Al-rich matrix with inclusions of euhedral Si crystals and of individual Si-spheres surrounded by a matrix of Al_2O_3 and tiny Si-particles.

In the reaction zone between Al and quartz (Fig. 12, zone 3) several metallic alloys were detected.

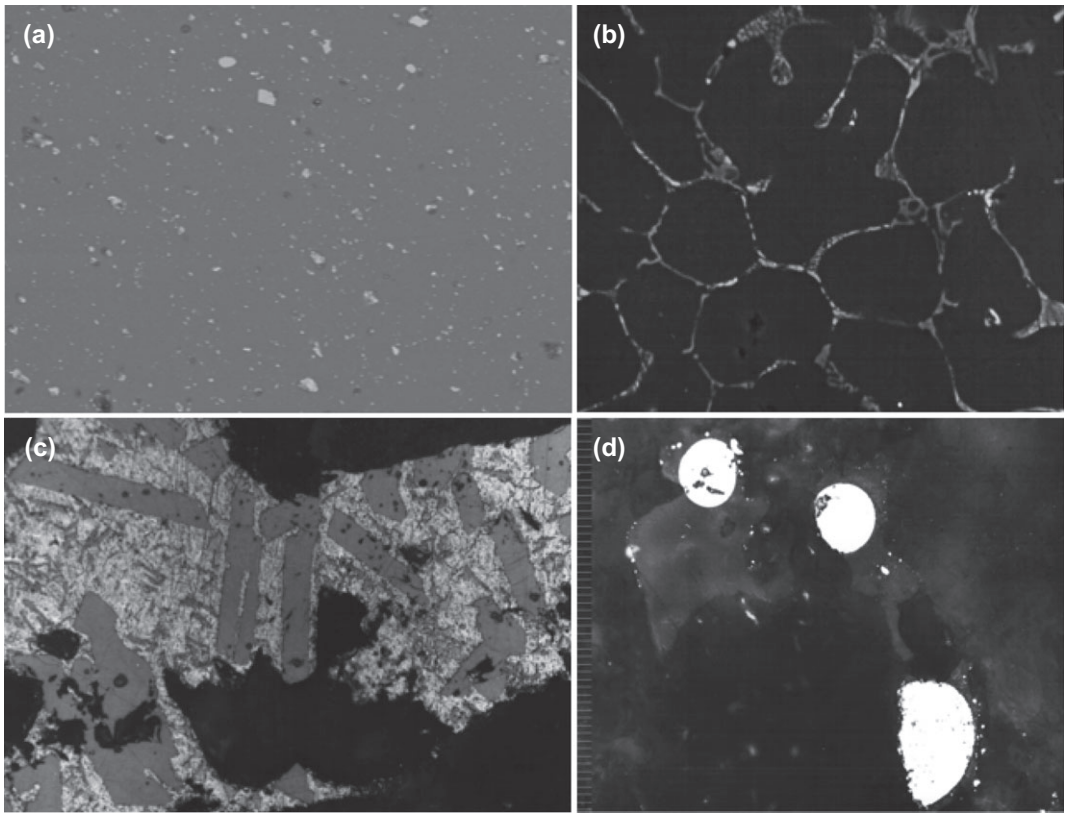


Fig. 14. SEM photos of the unshocked aluminum projectile containing "white" Al-poor and Cu-, Mn-, Fe-rich exsolved phases up to 10 μm in size (a) and shock-fused, recrystallized Al-crust on top of melt particles with network of eutectic phases (b); reflected light microscopic photos of gray euhedral Si crystals embedded in an Al-rich metal phase (speckled; c), and spheres of Si embedded in quartz glass (d); the latter three photos represent zone 1 (b) and zone 3 (c, d) of Fig. 12 and are taken from experiment No. 531; Cambridge Stereoscan, 20 kV (a, b); from Jammes (1984), (a) is a BSE image taken at 15 kV, courtesy of C. Hamann (Museum für Naturkunde, Berlin); widths of photos: 250 μm (a), 65 μm (b), 2.5 mm (c), 2.5 mm (d).

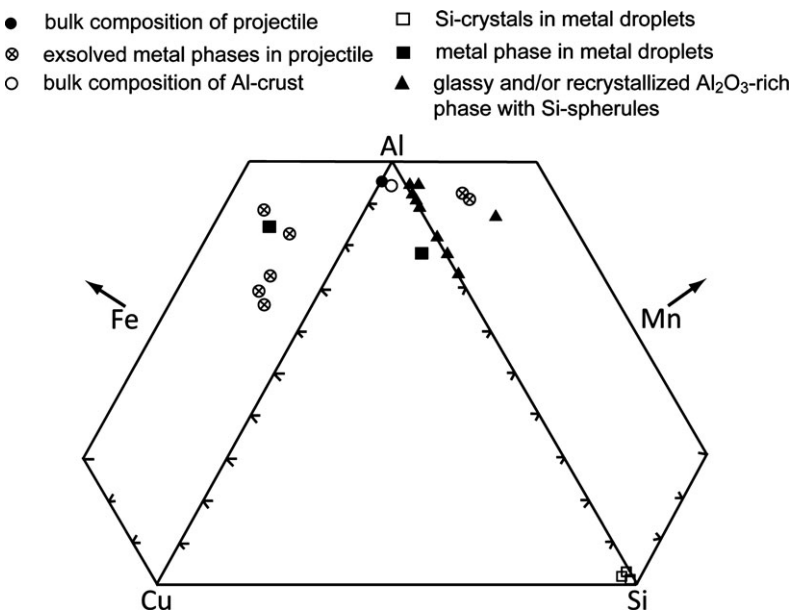


Fig. 15. Chemical composition of the aluminum projectile and of the reaction products formed at the interface of projectile and quartz sand; electron microprobe analyses obtained in melt particles of experiments Nos. 530, 532, and 533 at 15 kV and 0.1 μA with an electron beam diameter of 1 μm; from Jammes et al. (1983); for details see Hamann et al. (2016).

Most importantly are $CuAl_2$ (khatyrkite), a mineral first discovered in the CV3 chondrite Khatyrka (Bindi et al. 2012) and a Fe-Mn-rich phase $(Fe,Mn)_xSi(Al,Cu)_y$ (Hamann et al. 2016). The presence of khatyrkite in melt particles of Series II obviously represents the first experimental shock wave synthesis of this mineral. These phases indicate temperatures of at least 2000 °C due to the exothermic reaction Al and SiO_2 forming $Si + Al_2O_3$.

Shock Wave Attenuation and Zones of Shock Metamorphism

If the original color of the constituent quartz grains is preserved the above modal analyses can be utilized to reconstruct zones of decreasing shock pressure as a function of distance from a point, that is located at some depth (equivalent depth of burst which roughly corresponds to one projectile diameter, e.g., Melosh 1989) below the point of impact. The geometry of the expanding shock wave is often approximated by a hemisphere centered at a point some distance below the surface, so that the shock wave also travels upwards in the target and hits the surface from below (e.g., Melosh 1989).

The results are shown in Fig. 16 (right frame) by the dotted black lines >13 GPa and >0.9 GPa. Zone 1 ("melting") is inside the >13 GPa contour line and comminution occurs at or modestly beyond the 0.9 GPa contour; zones 2 and 3 ("lithification") occupy the volume between the two contour lines. The reconstruction of the zones of melting and shock lithification may be inaccurate to some degree, because the preimpact position of any shock-metamorphosed particle within its color-coded, 2 cm wide source ring is not known. Therefore, the dashed lines for the melt and lithification zones in Fig. 16 have issues of resolution and are only approximate.

In modeling it is a standard procedure to determine the attenuation of the shock wave. We use tracers that were initially located at the center of each computational cell to record the peak shock pressure they experienced during the passage of the shock wave and the excavation of the crater. As a good approximation it can be assumed that tracers have not been displaced before the passage of the shock wave, thus, the initial position of each tracer approximately corresponds to the place where the maximum shock pressure was achieved. Figure 16 shows the maximum shock pressure of each tracer at its initial location for

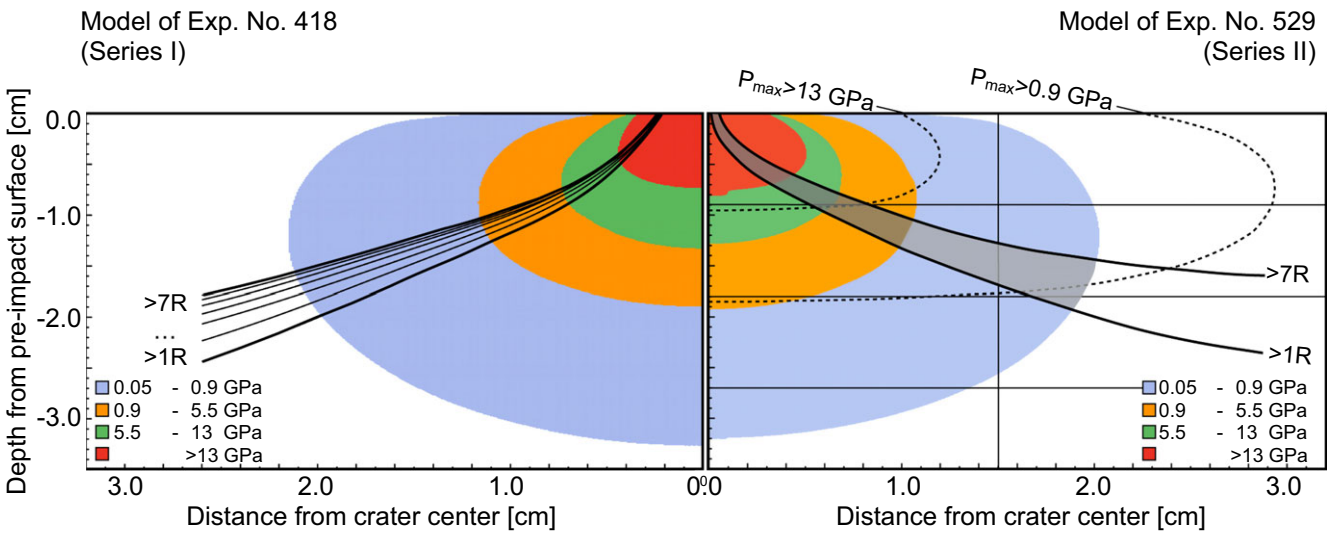


Fig. 16. Peak shock pressure Pmax as a function of distance from numerical modeling (color contours) of experiment No. 418 (Series I) and No. 529 (Series II). Pressures were recorded by tracers that are plotted at their initial position. The different color zones represent pressure ranges: (red) >13.0 GPa, (green) ~5.5 GPa–13.0 GPa, (orange) ~0.9 GPa–5.5 GPa, and (blue) 0.05–0.9 GPa. Right frame: the underlying grid represents the boundaries of the rings of colored sand (black sand = 0–1.5 cm and red sand 1.5–3.5 cm; see Fig. 3b); the overlaid dashed contours indicate zoning of peak shock pressure (Pmax) reconstructed from the ejecta distribution, the analysis of the shock melted and shock lithified particles, and from the color coding of grains preserved in some shock lithified aggregates. The outer contour represents the zone of shock lithification (Pmax>0.9 GPa), the inner contour the area where shock melting occurred (Pmax>13 GPa) (see also Stöffler et al. 1980a, 1983). Solid black lines mark the modeled excavation zone (<1R) and the zone of ejection distance >7R, respectively (see also Fig. 9, lower left frame). On the right frame the dark gray zone represents the volume of melt + lithified aggregates (>0.9 GPa) and the light gray zone represents the volume of comminuted grains (0.05–0.9 GPa) that were ejected out of the crater and deposited between 1 and 7 crater radii.

the model of experiment No. 418 (Series I, LEXAN) and experiment No. 529 (Series II, aluminum). The four different colors correspond to the four pressure ranges we were able to distinguish among the ejected quartz grains in the experiments and they are very similar in both simulations. According to the model results, the zone where melting occurs (zone 1, red, P_{max}>13 GPa) reaches down to a depth of 7 mm, which is in good agreement with the estimated depth of melting in the experiments (~9 mm; Fig. 16, right frame, inner dashed contour). Shock lithification (zone 2 and 3, green and orange, P_{max}>0.9 GPa) occurs in the models to a depth of 18 mm, which corresponds exactly to the experimental data (Fig. 16, right frame, light gray contour). All melt and aggregate particles (melt and shock lithified sand) originate from the upper two layers (0–18 mm depth), which corresponds to ~30% of the transient crater depth and slightly more than 50% of the excavation depth. However, in the experiments the lateral extent of the melt and lithification zone appears to be more than twice as wide as in the models (Fig. 16, right frame) and amounts to some 10% of the final crater diameter. The relatively imprecise reconstruction of the original location of shocked sand grains may explain the difference between model and experiments in terms of the radial attenuation of the shock wave.

The numerical models also quantify the absolute amount (= volume or mass) of sand that experienced the shock pressures of our four observed zones of shock metamorphism. Running very high-resolution models (60 CPPR—cells per projectile radius) that were stopped when the shock wave attenuated to pressures below ~10 GPa. Simulating experiment No. 529 we obtained 0.7 g originating in the red zone of Fig. 16, and 1.38 g (green zone), 5.8 g (orange zone), and 38.7 g (blue zone). Simulation No. 418 yielded 0.53 g, 1.84 g, 6.51 g, and 43.23 g, respectively. According to our models, a total of 46.6 g and 52.11 g of target material were shocked to >0.05 GPa in Series II and I, respectively. These totals compare to 74.4 g obtained from observations of Series I experiment No. 387. The calculated amount of melt and aggregate particles (>0.9 GPa) is 8.9 g for exp. No. 418), which is somewhat more than the 2.3 g determined from experiment No. 387 (see table 4 in Stöffler et al. 1975). Considering the somewhat larger crater volume of 1672 cm^3 in the model in comparison to 1467 cm^3 in the experiment (the value corresponds to experiment No. 387, which is very similar to No. 418) the percentage of melt and aggregate particles relative to the crater mass in model and experiment are 0.3% and 0.1%, respectively.

For Series II, No. 529, we determined in the model a total mass of melt and aggregate particles of 7.9 g. A comparison with the experimental value is difficult as the data scatter among the different experiments between 1 and 30 g. Omitting the largest outliers (exp. No. 532: 18 g; No. 533: 24 g, and No. 549: 30 g) we obtain an average value of 3.6 g. The reason for the large scatter in Series II is most likely due to having collected and analyzed only two neighboring or opposite sectors instead of all 360 degrees as in Series I experiment No. 387, which may have caused some bias if the offset between the point of impact and the geometric center of the sample pan was shifted along the axis of the sample sectors or perpendicular to it.

Mass Distribution of Shock Metamorphic Particles as a Function of Distance

For both cratering experiment Series I and II the mass distribution of the three types of shock metamorphic particles have been determined. However, the analysis of the data in particular for Series II revealed some inconsistencies and deviations from principles of ejection mechanics that most likely arose from (1) incomplete sampling (only two opposite or neighboring 90° sectors were analyzed in Series II), and (2) the offset between ground-zero and the geometric center of the color-coded rings. For this reason, we discuss here only the results from experiment No. 387 of Series I where all sectors were sampled, thus yielding more reliable results. Nevertheless, it should be noted here that the quantitative analysis of the distribution of shock metamorphic particles as a function of distance is error-prone due to the small volume of material relative to the total volume of ejecta that has experienced sufficiently high shock pressures. In principle, the same holds true for the number of tracers in the models. However, if we combine high-resolution models of the early stage of shock wave propagation (see the Shock Wave Attenuation and Zones of Shock Metamorphism section) with models of crater formation and ejection (see the Relationship Between Preimpact Location and Ejection Distance of Particles section) we were able to determine the distribution of shocked material in the models.

Melt Particles and Shock Lithified Aggregates

In Fig. 17 the mass distribution of melt particles and of shock lithified aggregates of sand is plotted as a function of horizontal radial distance from the point of impact and in the vertically placed catchers at different angles from the target surface (Series I). It should be noted here that inside the crater it was difficult to distinguish shock-metamorphosed particles that were ejected and fell back into the crater from displaced shock-metamorphosed particles that have never left

the crater. Beyond the crater rim the preimpact plane was marked by the lacquer that was sprayed on the surface making the identification of ejecta material unambiguous. The most important conclusions from the mass distributions of melt and shock lithified aggregates (Fig. 17) are the following. (1) The mass distribution of these particles is inverse to the mass distribution of the total ejecta, while the total ejecta mass decreases with distance, the modal content of "shocked" particles increases from the crater rim to about 7 crater radii and, with increasing angle in the vertical catchers. (2) Both types of shocked particles are highly concentrated inside the crater (2.5% of the total allochthonous material in experiment No. 387), and in the most distal ejecta, that is, in the two uppermost vertical catchers at angles between 35° and 50° from the ground-zero plane (1.5% of the total mass of ejecta in experiment No. 387). These characteristics generally also hold for cratering experiments of Series II (aluminum projectiles) although the trend is less obvious due to the large error as a result of incomplete sampling.

To quantify the abundance of melt and shock lithified aggregates (>0.9 GPa) in the models, we use the high-resolution models of shock wave propagation to determine the area where shock pressures >0.9 GPa occur (see Fig. 16, left frame) and the low-resolution models of the ejecta distribution to determine the zones of excavation for different ejection distances (Fig 16, left). To determine how much melt and shock lithified material is ejected to a certain distance, we calculated the volume of the solid of revolution bounded by the excavation zone contours for different ejection distances (1–7R). For example, the dark gray shaded area between the excavation zone contours 1R and 7R (Fig. 16, right frame) represents the total ejected melt and shock lithified material that lands inside the horizontal catcher range. The black horizontal bars in Fig. 17 indicate the model results for the abundance of agglutinary particles. The data agree well with the observed distribution for the Series I experiments, although we found a slightly larger amount of shocked particles at almost all distances except at 1–2R. This deviation is somewhat expected, as it is likely that some shocked particles have been overlooked during the analysis of the experiments. In turn, in the models the determination of the ejecta that land close to the crater rim (1–2R) may be inaccurate as the ballistic assumption may not be satisfied in this area.

For the model of experiment No. 418 we also determined the fraction of shocked particles that remain inside the crater: 0.38 g of melt (>13 GPa), 4 g of shock lithified grains (0.9–13 GPa), and 19.8 g of shock-comminuted grains (0.05–0.9 GPa), which corresponds to 0.015%, 0.15%, and 0.76% of the crater mass (2589 g). As these data were obtained from high-resolution models, we could not determine the final location of the material inside the crater. From low-resolution crater formation models we can infer that a small fraction of shocked material lines the wall of the transient crater. This material would possibly

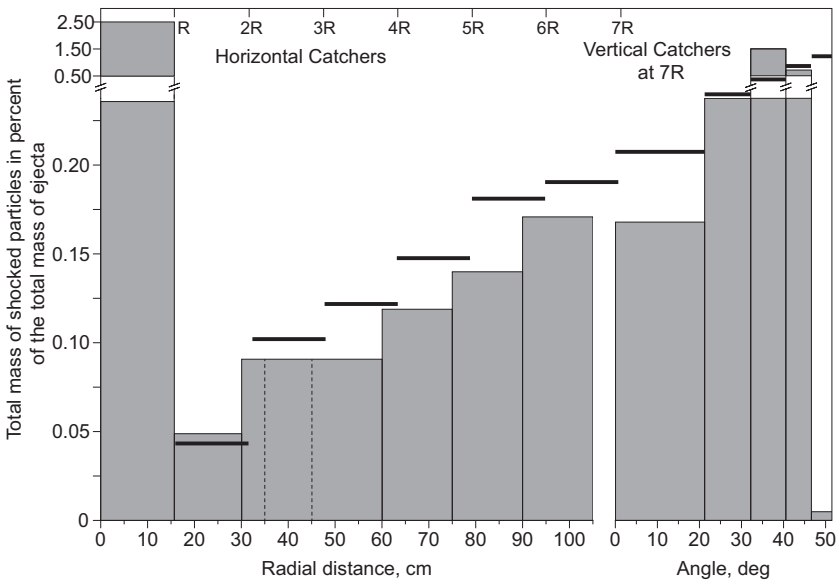


Fig. 17. Distribution of the percentage of the total mass of melt particles and shock lithified aggregates of sand normalized to the total amount of ejecta at each catcher for experiment No. 387 of Series I (gray columns) and numerical model of experiment No. 418 of Series I (black horizontal bars; length of bars indicate the range of bins for 1–7 radii of the modeled crater).

slump back and accumulate at the center of the final crater.

The large deviation between model and experiment at the upper most vertical catcher (46.9–51.5) clearly is due to a sampling error. The model shows a clear continuation of the trend with an increase in shocked material with distance, which is in agreement with the expected distribution of shocked material.

In the Series I experiment No. 387, which is nearly identical to No. 418 (Table 1a), 1.94 g of agglutinary particles were found inside the crater (see table 4 in Stöffler et al. 1975), which is about half of the mass determined in the model (4.38 g). This deviation may be a consequence that some of the shocked material was buried underneath some slump material from the crater wall and, thus, was lacking in the determination of the total mass of melt and shock lithified grains inside the crater. This would also explain why there is some deviation between model and experiments for the total amount of shocked material. It is not clear how much of the observed shocked material inside the crater represents fallback material or material that never left the crater. In the high-speed camera recording of the experiment No. 420, some minor incandescent material was traveling vertically upwards and fell back into the crater. It is clearly evident that a thin veneer of white powder-like comminuted sand covers the crater floor (see Fig. 4). The fallback of shocked (molten, lithified, and comminuted) material inside the crater can only be explained by nonballistic processes after formation of the transient cavity.

Shock-Comminuted Quartz

The mass distribution of the shock-comminuted sand (excess mass in the grain size fractions 0.063–0.125 mm and <0.063 mm compared to unshocked sand) will be discussed here only for the most completely sampled experiment, No. 387 of Series I (Fig. 18). For the interpretation of this graph, it should be kept in mind that about 94% of all grains of unshocked sand are in the 0.063–0.25 mm size fraction. The observed mass distribution of shock-comminuted sand is qualitatively similar to the distribution of melt and shock lithified particles shown in Fig. 17. For Series I experiment No. 418, the first maximum in the mass distribution of shock-comminuted sand is inside the crater and corresponds to more than 50% of the total mass of allochthonous material inside the crater (Fig. 18) compared to only 2.5% in the case of melt and shock lithified particles for Series I experiment No. 387 (Fig. 17). The second maximum is observed in the most distal ejecta and corresponds to about 2% of the total ejected mass compared to 1.6% in the case of melt and lithified particles (experiment No. 387).

We determined the distribution of comminuted sand grains (0.05–0.9 GPa) in the models according to the procedure described in the previous subsection (Melt

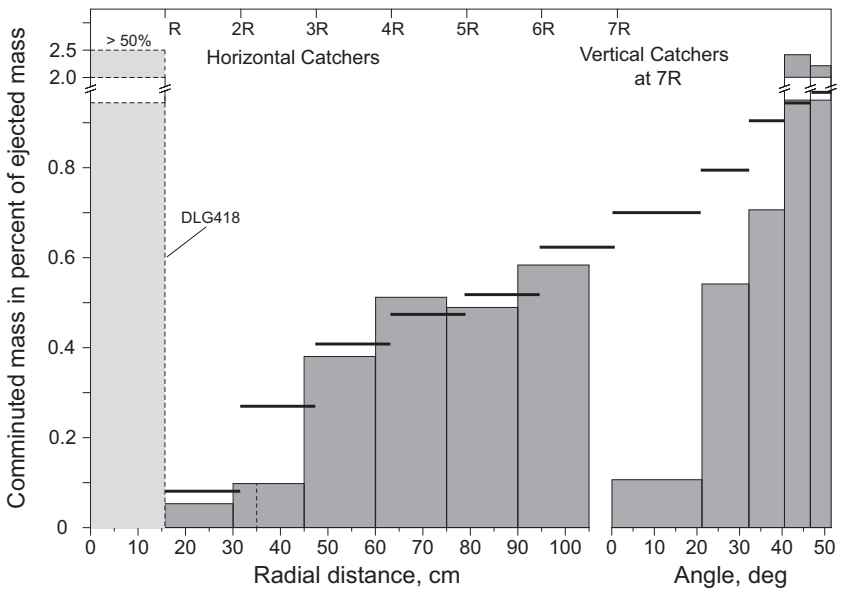


Fig. 18. Mass distribution of shock-comminuted sand (excess mass in the grain size fractions 0.063–0.125 mm and <0.063 mm compared to unshocked sand) for experiment No. 387 of Series I (gray columns) and numerical model of experiment No. 418 Series I (black horizontal bars, length of bars indicate the range of bins for 1–7 radii of the modeled crater); data for crater floor (light gray column) from experiment No. 418.

Particles and Shock Lithified Aggregates). Although the threshold pressures for the comminution of sand grains may be only approximately correct, we found surprisingly good agreement between models and observations (Fig. 18).

Effect of Agglutinary and Comminuted Particles on the Grain Size Distribution of Ejecta

Detailed data on the grain size distribution of ejecta as a function of radial distance from the point of impact were obtained for all Series II experiments (Table 1a). Six grain size fractions (<0.063, 0.063–0.125, 0.125–0.25, 0.25–0.5, 0.5–1.0, and >1.0 mm) were determined for six experiments (Nos. 529, 532, 534, and 548–550) and five fractions (<0.125 mm instead of <0.063 and 0.063–0.125 mm) for the remaining three experiments (Nos. 530, 531, 533). Based on these data, average weight percentages of all nine experiments were calculated for three grain size fractions (1) <0.125 mm, (2) 0.125–0.250 mm, and (3) >0.250 mm to be compared with the corresponding grain size fractions of the original quartz sand. The weight percentages in each of these fractions were plotted as a function of radial distance from the point of impact and as a function of the angle of the vertical catchers. The results are shown in Fig. 19.

Apparently, there is an excess mass in the two size fractions <0.125 mm and >0.250 mm over the original sand in all sampled ejecta, and consequently a mass deficiency in the size fraction 0.125–0.250 mm. This fraction represents some 57% of the mass of the unshocked sand. It is most distinctly reduced in the most distal ejecta where it achieves a minimum of 40%. It decreases slightly with increasing distance from the crater rim, that is, from ~56% to ~48% at 7 crater radii. On the other hand, the size fraction >0.250 mm (0.8% of the unshocked sand) shows a strong excess of mass in the vertical catchers at angles between 40° and 50° where it reaches a maximum of ~6.5%. As expected, the fine-grained fraction (size fraction <0.125) displays an excess mass compared to the unshocked sand in all sampled ejecta and has again a strong maximum in the high angle catchers (~56% compared ~42% of the unshocked sand).

Modeling of the size-frequency distribution of ejecta is currently not possible and we cannot provide any data for comparison here. It should also be noted that the observed comminution and agglutination of sand grains is dominated by the granular state of the target material, and a direct comparison with the size-frequency distribution of ejecta from coherent rock (e.g., Buhl et al. 2014) is not possible.

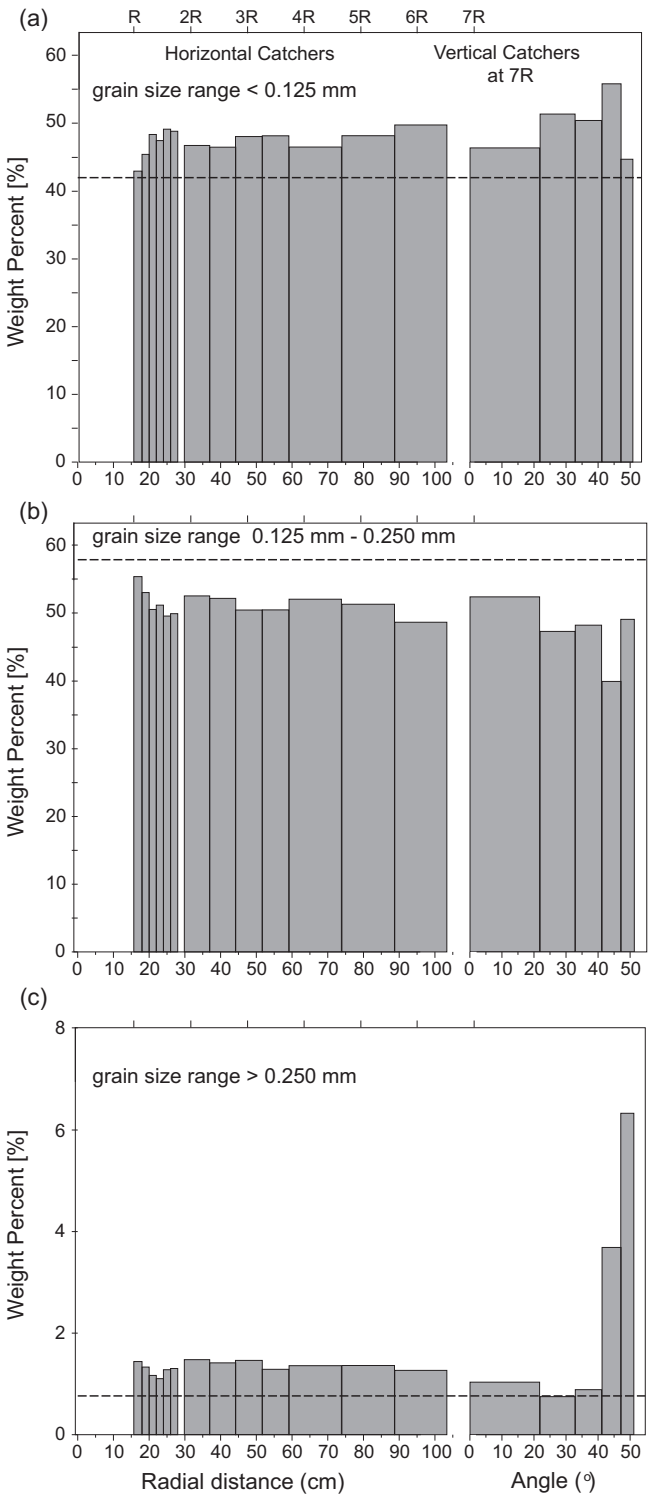


Fig. 19. Relative masses (weight percent) of ejecta in three grain size fractions (a, b, c). The mass fractions in each sampled ring or vertical catcher represent averages calculated from the data of nine cratering experiments with Al-projectiles (Nos. 529, 530, 531, 532, 533, 534, 548, 549, and 550). The horizontal dashed lines mark the original fraction of plain sand.

COMPARISON OF EXPERIMENTS AND NUMERICAL MODELING

Generally, the results from laboratory experiments and numerical modeling are in good agreement and confirm some basic understanding of crater formation, the ejection of shocked material, and the deposition of the ejecta blanket. The study demonstrates that modeling and experiments are complementary methodological approaches providing insights into the cratering and ejection processes on different scales. The models are capable of reproducing such macroscale observations as crater size and the thickness of the ejecta blanket as a function of distance sufficiently accurately. Deviations between the models and the experiments with regard to the place of origin of ejected material are attributed to the relatively poor resolution of the experiments (absolute size of the colored rings and catcher bins), not ideal symmetry (offset between impact and geometric center of colored rings), and the potential error in recognizing the specific color of individual grains. Nevertheless, the models still roughly match the observations and, therefore, allow for confident, but somewhat idealized predictions of the distribution of different lithologies in an ejecta blanket for impacts into a stratigraphically well-defined target.

The experiments provide further insights into the shock modification and its effects on grain size distribution as a function of radial distance from the crater that can only be partially resolved and reproduced by the models. The experiments confirm the recently reduced shock pressures required to generate melting (Kowitz et al. 2013) as a consequence of porosity (Wünnemann et al. 2008).

In conclusion, this study confirms the generally hypothesized distribution of shock-metamorphosed particles and their place of origin in the target. The increase of the degree of shock metamorphism of ejecta with radial range is based on theoretical assumptions and numerical models, but has not been quantified in experiments so far. Validation of models is crucial for further more general studies on ejecta distribution on arbitrary dimensional scales, higher impact velocities, varying impact angles, different gravitational fields, and a wide range of target properties (strength, porosity).

LIMITATIONS OF METHODOLOGICAL APPROACHES

In comparison to natural crater formation and ejection of material on planetary surfaces, laboratory experiments simplify the real conditions. In addition to different size-scales, other parameters such as the impact angle (only vertical impacts were considered) or the impact velocity (which is in the experiments only 1/2–1/3 of the actual average impact velocity on Earth and Moon) affect the cratering process although the latter restriction does not hold for asteroids. Moreover, the sand target in the experiments behaves like a granular material that may only be representative for planetary regoliths, and not like cohesive rocks that are more typical for large-scale cratering on planetary targets, although it is well known that strength needs to be scaled (reduced) relative to the size of an experiment. To counter these limitations, impacts are known to resemble in many respects explosion processes that can be approximated by a point source if the impact velocity exceeds the speed of sound of the target material (Oberbeck 1971), a precondition our experiments clearly satisfy. Therefore, we do not expect that the significantly lower impact velocities in the experiments than in nature lead to a different comprehension of the basic process. The impact-explosion analogy also justifies the neglect of the influence of the impact angle to some extent; however, it is well known that the vast majority of impacts occur at an oblique angle of incidence, with the most likely impact angle of 45°, which explains the asymmetric distribution of ejecta around natural craters (e.g., Gault and Wedekind 1978; Herrick and Forsberg-Tylor 2003; Anderson et al. 2004). Our study does not address the effect of the impact angle on ejecta distribution, but rigorously validated models enable us to extend the results of this study to oblique impact scenarios. Finally, the granular rheology of the sand targets may be considered as a good approximation of (1) the heavily shattered state of, for example, the lunar megaregolith or asteroidal rubble piles and (2) for the surficial regolith layer present on all atmosphereless bodies, for example, the Moon, Mercury, and asteroids. Sand targets have been used as laboratory analog material for cratering on planetary surfaces in numerous previous studies (e.g., Gault and Heitowit 1963; Gault et al. 1968; Schmidt and Housen 1987; Hermalyn and Schultz 2011).

Next to the intrinsic limitations of the experiments to resemble nature, they also suffer from some technical problems that may have caused some uncertainty in the results. Although the experiments were intended to be radially symmetric, the results clearly deviate from the idealized condition. As mentioned above, we identified the main reason for this to be the offset of the aim point and actual impact site relative to the geometric center of the rings of colored sand. Other explanations for the asymmetric distribution of ejecta in the experiments may be (1) the interaction of ejected sand grains with the gas flow in the wake of the projectile and (2) with the perturbed low-density atmosphere in the target chamber (ambient pressure after evacuation was 67–400 Pa). The latter can be estimated by calculating the atmospheric

drag on ejected particles for the size range of recovered grains. The launch velocity for the ejecta under consideration is ~3.3 m s^{-1} for a deposition distance of 100 cm. The smallest particles (0.063 mm) traverse through an atmosphere of 400 Pa to a distance of ~109 cm if ejected at 3.3 m s^{-1} at an angle of 45°. If no atmosphere is present the same particle lands 2 cm farther away. The difference is less than 2%. This simple sample calculation demonstrates that atmospheric effects were negligible in the experiments.

Another previously mentioned problem in the analysis of the ejecta distribution in the experiments is given by the fact that not all sand grains may have retained their original color during the cratering process. Due to the irregular shape of the sand grains, all of their surfaces were not entirely covered with dye initially, and some of the color may have rubbed off during the ejection process, such that they were counted as plain grains. The problem appears to be more pronounced in Series II as the results of Series I published in Stöffler et al. (1975) are much more consistent with the expected ejecta distribution of shocked particles. Also some grains were unintentionally dyed black by the lacquer that was sprayed on the target to mark the preimpact surface. Presumably, the grains at the surface ejected to all distances were erroneously interpreted to originate all from the innermost (black) region.

The numerical results suffer from the fact that not all processes are resolved sufficiently. In the models of crater formation and ejecta distribution only one tracer represents all sand grains in a radial space increment of 0.395 mm width, which makes the number of ejected tracers that are deposited at a given distance very small. The resolution is much better in the models to determine the amount of shocked material, but those models cover only a short, very early stage of crater formation and do not allow for the reconstruction of the ejecta distribution. The geometric estimates of the amount of ejected, highly shocked material (see Fig. 16) has to be considered as first order approximation relating the zones of different levels of shock pressure with zones of ejection distances.

Our approach using tracers to model the ejecta distribution does not account for the size-frequency distribution resulting from melting, aggregation, and comminution of grains in the ejecta. We consider the flight of ejecta as ballistic and the atmospheric drag acting on even the smallest particles to be negligible, such that the size distribution of ejecta does not affect their distribution pattern. However, the size-frequency distribution of ejecta in natural impacts is poorly known and small dust particles can strongly interact with an atmosphere and the plume of vaporized material. Although validating models of the ejecta plume is highly desirable, it is not the subject of this study. In particular, on atmosphereless planetary objects, the vast majority of ejecta are deposited ballistically making our approach directly applicable to impacts on, for example, the Moon or asteroids.

POTENTIAL APPLICATION OF THE RESULTS TO NATURAL IMPACT PROCESSES

Our complementary approach of numerical modeling and laboratory experiments provides insights into the shock history, ejection, and emplacement of debris expelled from impact craters and provides confidence in modeling of these processes. Our results contribute to a better understanding of the ejection and distribution of impactites (e.g., melt and shock metamorphic particles) originating from some given point in the preimpact target, which is in particular important to quantify the so-called impact gardening on planetary surfaces, a poorly quantified surface process. For example, on the Moon, the bombardment of the near-surface regolith is dominated by the smallest, most frequent impacts. Complementary to scaling laws (e.g., Housen et al. 1983; Housen and Holsapple 2011) to account for the effect of varying gravity on different planets, the presented results are comparable to the formation of small impact craters and the ejection of material on atmosphereless planetary bodies such as the Moon, Mercury, and asteroids. Additional modeling seems on firm grounds to address different collisional settings, such as varying composition, porosity, and strength of the target; planetary gravity; and impact velocities. Following first attempts by Collins and Wünnemann (2007), a systematic parameter study using a suite of "numerical experiments" should be the next step to improve existing scaling laws with regard to the effect of material properties such as strength, friction, and porosity. In addition, models also allow for the reconstruction of the formation of large complex craters and impact basins, where scaling laws may fail to predict the outcome of such large impact events sufficiently accurately. However, the larger impacts, although less frequent, should dominate, and produce most of the highly shocked material (impact melt) in the near-surface strata and they are responsible for the vertical and lateral redistribution of substantial crustal and possibly mantle materials. Although our models in principle also allow for up-scaling our results to any arbitrary size scale (e.g., basin-size impacts) and impact velocity, the emplacement and mixing with near-surface strata needs to be further studied. With increasing impactor size, the amount of high-velocity, highly

shocked ejecta increases relative to the total amount of ejecta. As the launch velocity of ejecta corresponds to the deposition velocity, the mixing with local material upon landing becomes important for high-velocity, highly shocked, distal ejecta. The entrainment of local material can significantly contribute to the formation of ejecta deposits and the reworking of the surface strata. The process is commonly referred to as secondary mass wasting or ballistic sedimentation (Oberbeck 1975) and has only been quantified at the well-preserved ejecta deposits of the Ries crater in Germany where the proportion of locally entrained material increases with distance, reaching up to 90–95% of the total deposited mass at 2 to 3 crater radii (Hörz et al. 1983). First attempts to quantify secondary mass wasting by modeling based on the present study have been proposed in Zhu and Wünnemann (2013), but a better quantification requires knowledge of the size-frequency distribution of ejected debris. Currently, it is not possible to accurately relate the size and number of ejecta fragments to their ejection velocity, the impactor size and its velocity, and the material properties of the target body. Only observational data from terrestrial craters (megablocks at Ries; Sturm et al. 2015) and the distribution of boulders on the Moon (Moore 1971; Bart and Melosh 2010), empirical relationships (Vickery 1986, 1987; Buhl et al. 2014), or Weibull power-law distribution have been used in modeling attempts (Artemieva and Morgan 2009; Artemieva et al. 2013). A quantitative model of the size-frequency distribution of ejecta is the last missing piece for a more comprehensive and quantitative understanding of impact gardening. If this lacking information is added, our results may serve as a basis for modeling the formation, composition, and growth of regolith and subregolith layers on atmosphereless planetary bodies such as the Moon, if we start with a substrate of magmatic rocks (mare basalt or highland anorthosite) and make assumptions about the impactor flux. Because both small and large impact events produce significant effects, the problem is not trivial: a surface veneer develops on relatively short timescales, but is periodically overturned by larger impact events. The excavation, ejection, and deposition of material originating from different lithological and rheological units (depending on the preimpact structure of the target and the size of the impact event) forms a complex stratigraphy of impactites from numerous events (Heskin et al. 2002) representing the present day accessible regions on the Moon from where samples have been returned and dated (e.g., Stöffler et al. 2006). The reconstruction of the entire emplacement history and in particular, the shock exposure of lunar samples that have been interpreted as primary ejecta from certain basin-forming impact events, is essential for a better understanding of the chronology of the lunar crater record and, thus, the bombardment history of the inner planetary system.

The verification of the numerical models of the distribution of ejecta of different degree of shock metamorphism with the results from the laboratory experiments opens the perspective to simulate the regolith accumulation, redistribution, and melt content, which would allow us to estimate what abundances of melt with different ages could be expected in surface samples.

Acknowledgments—We are very grateful to Fred Hörz and Boris Ivanov for their constructive and insightful reviews that greatly improved the manuscript. We also thank Boris Ivanov, Jay Melosh, Gareth Collins, Tom Davison, and Dirk Elbeshausen for their help in developing iSALE (www.isalecode.de). This work was funded by DFG Grant# WU355/6-2. MHZ was partly supported by the Science and Technology Development Fund of Macau (075/2014/A2).

Editorial Handling—Dr. Natalia Artemieva

REFERENCES

Amsden A. A., Ruppel H. M., and Hirt C. W. 1980. *SALE: A simplified ALE computer program for fluid flow at all speeds*. Los Alamos, New Mexico: Los Alamos National Laboratory Report LA-8095. 101 p.

Anderson J. L. B., Schultz P. H., and Heineck J. T. 2004. Experimental ejection angles for oblique impacts: Implications for the subsurface flow-field. *Meteoritics & Planetary Science* 39:303–320.

Andrews R. J. 1977. Characteristics of debris from small-scale cratering experiments. In *Impact and explosion cratering: Planetary and terrestrial implications*, edited by Roddy D. J., Pepin R. O., and Merrill R. B. New York: Pergamon Press. pp. 1089–1100.

Artemieva N. A. and Morgan J. 2009. Modeling the formation of the K-Pg boundary layer. *Icarus* 201:768–780.

Artemieva N. A., Wünnemann K., Krien F., Reimold W. U., and Stöffler D. 2013. Ries crater and suevite revisited—Observations and modeling, Part II: Modeling. *Meteoritics & Planetary Science* 48:590–627.

Bart G. D. and Melosh H. J. 2010. Distribution of boulders ejected from lunar carters. *Icarus* 209:337–257.

Bindi L., Eiler J. M., Yunbin G., Hollister L. S., McPherson G. J., Steinhartdt P. J., and Yao N. 2012. Evidence for the extraterrestrial origin of a natural quasicrystal. *Proceedings of the National Academy of Sciences* 109:1396–1401.

Buhl E., Sommer F., Poelchau M. H., Dresen G., and Kenkmann T. 2014. Ejecta from experimental impact craters: Particle size distribution and fragmentation energy. *Icarus* 237:131–142.

Carlson R. H. and Jones G. D. 1965. Distribution of ejecta from cratering explosions in soils. *Journal of Geophysical Research* 70:1897–1910.

Cintala M. J., Berthoud L., and Hörz F. 1999. Ejection-velocity distributions from impacts into coarse-grained sand. *Meteoritics & Planetary Science* 34:605–623.

Collins G. S. 2014. Numerical simulations of impact crater formation with dilatancy. *Journal of Geophysical Research—Planets* 119:2600–2619. doi:10.1002/2014JE004708.

Collins G. S. and Wünnemann K. 2007. Numerical modeling of impact ejection processes in porous targets (abstract #1338). 38th Lunar and Planetary Science Conference. CD-ROM.

Collins G. S., Melosh H. J., and Ivanov B. A. 2004. Modeling damage and deformation in impact simulations. *Meteoritics & Planetary Science* 39:217–231.

Collins G. S., Kenkmann T., Osinski G. R., and Wünnemann K. 2008. Mid-sized complex crater collapse formation in mixed crystalline-sedimentary targets: Insight from modeling and observation. *Meteoritics & Planetary Science* 43:1955–1977.

Collins G. S., Melosh H. J., and Wünnemann K. 2011. Improvements to the ε-a porous compaction model for simulating impacts into high-porosity solar system objects. *International Journal of Impact Engineering* 38:434–439.

Davison T. M., Collins G. S., Elbeshausen D., Wünnemann K., and Kearsley A. T. 2011. Numerical modeling of oblique hypervelocity impacts on strong ductile targets. *Meteoritics & Planetary Science* 46:1510–1524.

Gault D. E. and Heitowit E. D. 1963. The partition of energy for hypervelocity impact craters formed in rock. In *Proceeding of the 6th Hypervelocity Impact Symposium, vol. 2*. Cleveland, Ohio: Firestone Rubber Company. pp. 419–456.

Gault D. E. and Wedekind J. A. 1978. Experimental studies of oblique impact. Proceedings, 9th Lunar and Planetary Science Conference. pp. 3843–3875.

Gault D. E., Quaide W. L., and Oberbeck V. R. 1968. Impact cratering mechanics and structures. In *Shock metamorphism of natural materials*, edited by French B. M. and Short N. M.. Baltimore, Maryland: Mono. pp. 87–100.

Güldemeister N., Wünnemann K., Durr K., and Hiermaier S. 2013. Propagation of impact-induced shock waves in porous sandstone using mesoscale modeling. *Meteoritics & Planetary Science* 48:115–133.

Güldemeister N., Wünnemann K., and Poelchau M. H. 2015. Scaling impact crater dimensions in cohesive rock by numerical modeling and laboratory experiments. *Geological Society of America Special Papers* 518:17–29. doi:10.1130/2015.2518(02).

Hamann C., Stöffler D., and Reimold W. U. 2016. Interaction of aluminum projectiles with quartz sand in impact experiments: Formation of khatyrkite ($CuAl_2$) and reduction of SiO_2 to Si. *Geochimica et Cosmochimica Acta* In press.

Heineck J. T., Schultz P. H., and Anderson J. L. B. 2002. Application of three-component PIV to the measurement of hypervelocity impact ejecta. *Journal of Visualization* 5:233–241.

Hermalyn B. and Schultz P. H. 2010. Early-stage ejecta velocity distribution for vertical hypervelocity impact into sand. *Icarus* 209:866–870. doi:10.1016/j.icarus.2010.05.025.

Hermalyn B. and Schultz P. H. 2011. Time-resolved studies of hypervelocity vertical impacts into porous particulate targets: Effects of projectile density on early-time coupling and crater growth. *Icarus* 216:269–279. doi:10.1016/j.icarus.2011.09.008.

Herrick R. R. and Forsberg-Tylor N. K. 2003. The shape and appearance of craters formed by oblique impact on the Moon and Venus. *Meteoritics & Planetary Science* 38:1551–1578.

Heskin L. A., Moss B. E., and McKinnon W. B. 2002. On estimating contributions of basin ejecta to regolith deposits at lunar sites. *Meteoritics & Planetary Science* 38:13–33.

Hörz F., Ostertag R., and Rainey D. A. 1983. Bunte Breccia of the Ries: Continuous deposits of large impact craters. *Review of Geophysics and Space Physics* 21:1667–1725.

Housen K. R. and Holsapple K. A. 2011. Ejecta from impact craters. *Icarus* 211:856–875.

Housen K. R., Schmidt R. M., and Holsapple K. A. 1983. Crater ejecta scaling laws—Fundamental forms based on dimensional anaylsis. *Journal of Geophysical Research* 88:2485–2499.

Ivanov B. A., Deniem D., and Neukum G. 1997. Implementation of dynamic strength models into 2D hydrocodes: Applications for atmospheric breakup and impact cratering. *International Journal of Impact Engineering* 20:411–430.

Jammes C. 1984. Experimentelle Stoßwellenmetamorphose von Quarzsand durch Hochgeschwindigkeits-Impakt von Aluminiumprojektilen. Diploma thesis. University of Münster, Germany. 62 p.

Jammes C., Stöffler D., Bischoff A., Reimold W. U., and Gault D. E. 1983. Reduction of SiO_2 to Si and metallurgical transformations in Al by hypervelocity impact of Al-projectile into quartz sand. 16th Lunar and Planetary Science Conference. pp. 347–348.

Johnson B. C., Bowling T. J., and Melosh H. J. 2014. Jetting during vertical impacts of spherical projectiles. *Icarus* 238:13–22.

Kieffer S. W. 1971. Shock metamorphism of the Coconino sandstone at Meteor Crater, Arizona. *Journal of Geophysical Research* 76:5449–5473.

Kieffer S. W., Phakey P. P., and Christie J. M. 1976. Shock processes in porous quartzite: Transmission electron microscope observations and theory. *Contributions to Mineralogy and Petrology* 59:41–93.

Kowitz A., Güldemeister N., Reimold W. U., Schmitt R. T., and Wünnemann K. 2013. Diaplectic quartz glass and SiO_2 melt experimentally generated at only 5 GPa shock pressure in porous sandstone: Laboratory observations and meso-scale numerical modeling. *Earth and Planetary Science Letters* 384:17–26.

McGetchin T., Settle M., and Head J. W. 1973. Radial thickness variation in impact crater ejecta, implication for lunar basin deposits. *Earth and Planetary Science Letters* 20:226–236. doi:10.1016/0012-821X (73)90162-3.

Melosh H. J. 1989. *Impact cratering: A geologic process*. New York: Oxford University Press. 245 p.

Melosh H. J. 2007. Hydrocode equation of state for SiO_2. *Meteoritics & Planetary Science* 42:2035–2182.

Melosh H. J., Ryan E. V., and Asphaug E. 1992. Dynamic fragmentation in impacts: Hydrocode simulation of laboratory impacts. *Journal of Geophysical Research* 97:14,735–14,759.

Moore H. J. 1971. *Large blocks around lunar craters, analysis of Apollo 10 photography and visual observations*.

Washington, D.C.: NASA SP-232, National Aeronautics and Space Administration. 232 p.

Oberbeck V. R. 1971. Laboratory simulation of impact cratering with high explosives. *Journal of Geophysical Research* 76:5732–5749.

Oberbeck V. R. 1975. The role of ballistic erosion and sedimentation in lunar stratigraphy. *Reviews of Geophysics and Space Physics* 13:337–362.

Ormö J., Melero-Asensio I., Housen K., Wünnemann K, Elbeshausen D., and Collins G. S. 2015. Scaling and reproducibility of craters produced at the Experimental Projectile Impact Chamber (EPIC), Centro de Astrobiología, Spain. *Meteoritics & Planetary Science* 50:2067–2086. doi:10.1111/maps.12560.

Pierazzo E., Artemieva N. A., Asphaug E., Baldwin E. C., Cazamias J., Coker R. F., Collins G. S., Crawford D. A., Davison T. M., Elbeshausen D., Holsapple K. A., Housen K. R., Korycansky D. G., and Wünnemann K. 2008. Validation of numerical codes for impact explosion cratering: Impacts on strength less and metal targets. *Meteoritics & Planetary Science* 43:1917–1938.

Schmidt R. M. and Housen K. R. 1987. Some recent advances in the scaling of impact and explosion cratering. *International Journal of Impact Engineering* 5:543–560.

Stöffler D. and Grieve R. A. F. 2007. Impactites—A proposal on behalf of the IUGS subcommission on the systematics of metamorphic rocks. In *Metamorphic rocks: A classification and glossary of terms, recommendations of the International Union of Geological Sciences*, edited by Fettes D. and Desmons J. Cambridge, UK: Cambridge University Press. pp. 82–92.

Stöffler D., Gault D. E., Wedekind J., and Polkowski G. 1975. Experimental hypervelocity impact into quartz sand: Distribution and shock metamorphism of ejecta. *Journal of Geophysical Research* 80:4062–4077.

Stöffler D., Gault D. E., and Reimold W. U. 1980a. Cratering experiments in noncohesive and weakly cohesive sand: Excavation mode and ejecta characteristics (abstract). *Conference on multi-ring basins, formation and evolution*. LPI Contribution 414. Houston, Texas: The Lunar and Planetary Institute. pp. 89–91.

Stöffler D., Gault D. E., and Reimold W. U. 1980b. Experimental hypervelocity impact into quartz sand: Pre-impact location of ejecta. *Meteoritics* 15:371–372.

Stöffler D., Reimold W. U., Eylert D., Wiedenhues R., Bachmann G., Pankrath R., Ostertag R., Malley J., and Jammes C. 1983. Die experimentelle Erzeugung von Impaktkratern in geologischen Materialien durch Hochgeschwindigkeitsprojektile. Final Report to the Ministry of Science and Research of the State of Nordrhein-Westfalen.

Stöffler D., Ryder G., Artemieva N. A., Cintala M. J., Grieve R. A. F., and Ivanov B. A. 2006. Cratering history and lunar chronology. In *New views of the Moon*, edited by Joliff B. L, Wieczorek M. A., Shearer C. K., and Neal C. R. Reviews in Mineralogy and Geochemistry, vol. 60. Washington, D.C.: The Mineralogical Society of America. pp. 519–596.

Stöffler D., Reimold W. U., Jacob J., Hansen B. K., Summerson I. A. T., Artemieva N. A., and Wünnemann K. 2013. Ries crater and suevite revisited - Observations and modeling, Part I: Observations. *Meteoritics & Planetary Science* 48:515–589.

Sturm S., Kenkmann T., Willmes M., Pösges G., and Hiesinger H. 2015. The distribution of megablocks in the Ries crater, Germany: Remote sensing, field investigation, and statistical analyses. *Meteoritics & Planetary Science* 50:141–171.

Thompson S. L. and Lauson H. S. 1972. Improvements in the CHART D radiation-hydrodynamic code III: Revised analytic equations of state. Sandia National Laboratory Report SC-RR-71 0714. 113 p.

Vickery A. M. 1986. Size-velocity distribution of large ejecta fragments. *Icarus* 67:224–236. doi:10.1016/0019-1035(86)90105-3.

Vickery A. M. 1987. Variation in ejecta size with ejection velocity. *Geophysical Research Letters* 14:726–729. doi:10.1029/GL014i007p00726.

Wada K., Senshu H., and Matsui T. 2006. Numerical simulation of impact cratering on granular material. *Icarus* 180:528–545.

Wünnemann K., Collins G. S., and Melosh H. J. 2006. A strain-based porosity model for use in hydrocode simulations of impacts and implications for transient-crater growth in porous targets. *Icarus* 180:514–527.

Wünneman K., Collins G. S., and Osinski G. R. 2008. Numerical modelling of impact melt production in porous rocks. *Earth and Planetary Science Letters* 269: 529–538.

Wünnemann K., Nowka D., Collins G. S., Elbeshausen D., and Bierhaus M. 2011. Scaling of impact crater formation on planetary surfaces—Insights from numerical modeling. *Proceedings of the 11th Hypervelocity Impact Symposium*, edited by Schäfer F. and Hiermaier S. Schriftenreihe Forschungsergebnisse aus der Kurzzeitdynamik, vol. 20. Freiburg: Fraunhofer EMI. pp. 1–16.

Zhu M.-H. and Wünnemann K. 2013. Modeling of meteorite impact-induced secondary mass wasting—Case study by means of the Bunte Breccia ejecta blanket at Ries crater, Germany (abstract #1921). 44th Lunar and Planetary Science Conference. CD-ROM.

Zhu M.-H., Wünnemann K., and Potter R. W. K. 2015. Numerical modeling of the ejecta distribution and formation of the Orientale basin on the Moon. *Journal of Geophysical Research—Planets* 120: doi:10.1002/2015JE004827.